

게임엔진 기초

Godot Animation Blending

강영민

동명대학교 게임그래픽학과



https://www.youtube.com/watch?v=j48WTY1w_Po

- 필요한 도구

블렌더: FBX 파일 통합

Mixamo 서비스: 애니메이션 데이터 다운로드

블렌더 설치

<https://builder.blender.org/download/daily/>

[Developer](#) [Projects](#) [Docs](#) [Blog](#) [Forum](#) **[Builds](#)**

[Daily](#) [Branch](#) [Patch](#) [Python Module](#) [Dashboard](#) [Tests](#) [Buildbot](#) [Latest Changes](#)

Daily Builds

The following builds have the latest features and bug fixes, but they can be unstable and break your files. **Use at your own risk.**

Windows

macOS

Linux

All Architectures

Version	Variant	Reference	SHA	Date	Architecture	
Blender 5.1.0	Alpha	616bdf070426	SHA	05 Nov 02:07	arm64	Download
Blender 5.1.0	Alpha	616bdf070426	SHA	05 Nov 01:50	x64	Download
Blender 5.0.0	Beta	1f5254094830	SHA	05 Nov 02:09	arm64	Download
Blender 5.0.0	Beta	08de9aeb4348	SHA	04 Nov 13:48	x64	Download

[All Archived Builds](#)

Do your part!

Support Blender developers to work on:

- Bug fixes and stability
- 3 releases a year
- New features such as [Vulkan](#), [improved UV tools](#), [NPR](#), and more.

Every donation counts. Thank you!

Monthly

One-time

€ 25

€ 50

€ 100

€ 150

€ 250

€ 500

Donate, just once

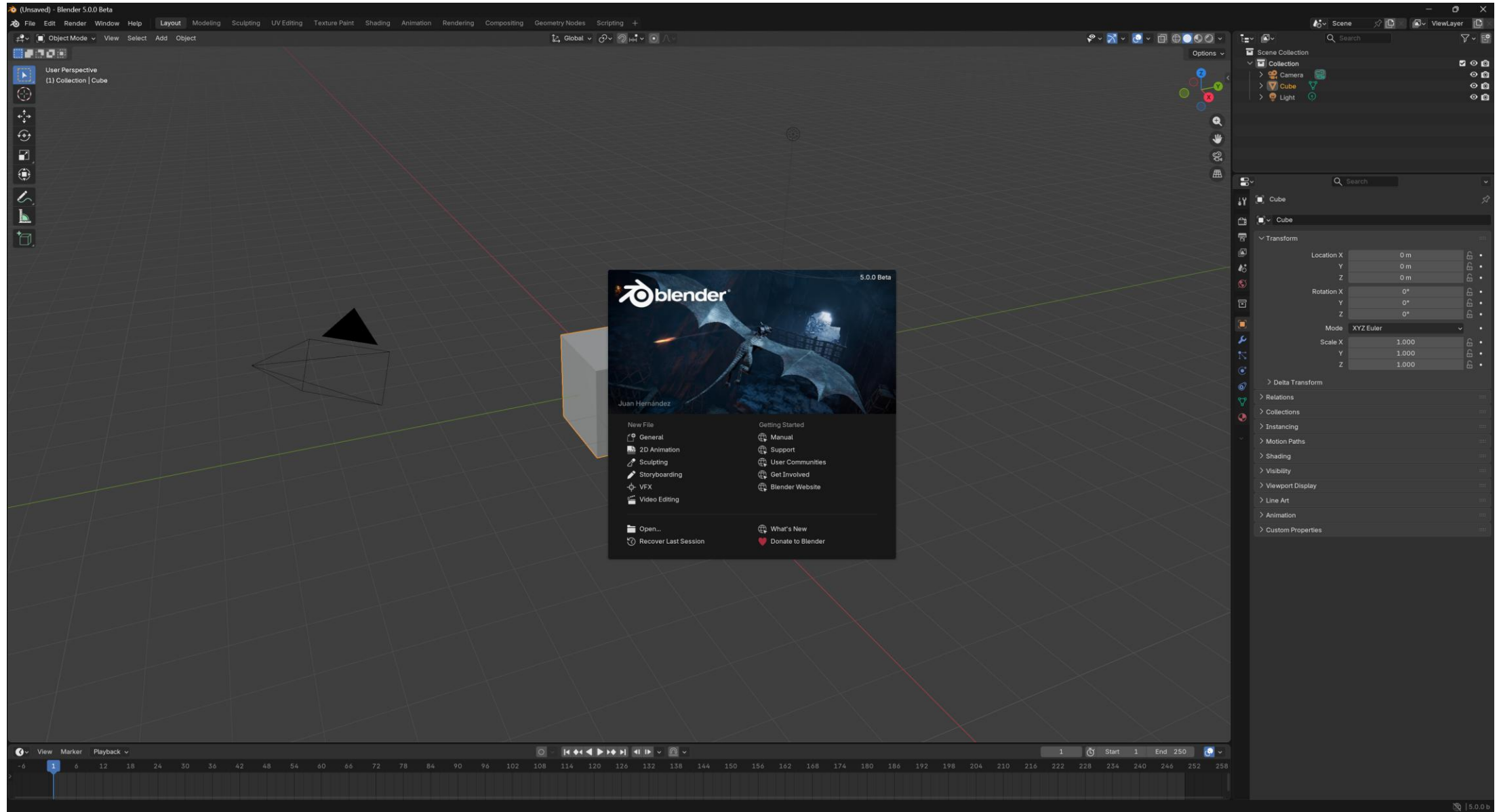
€ 100

Donate

[Corporate memberships](#)

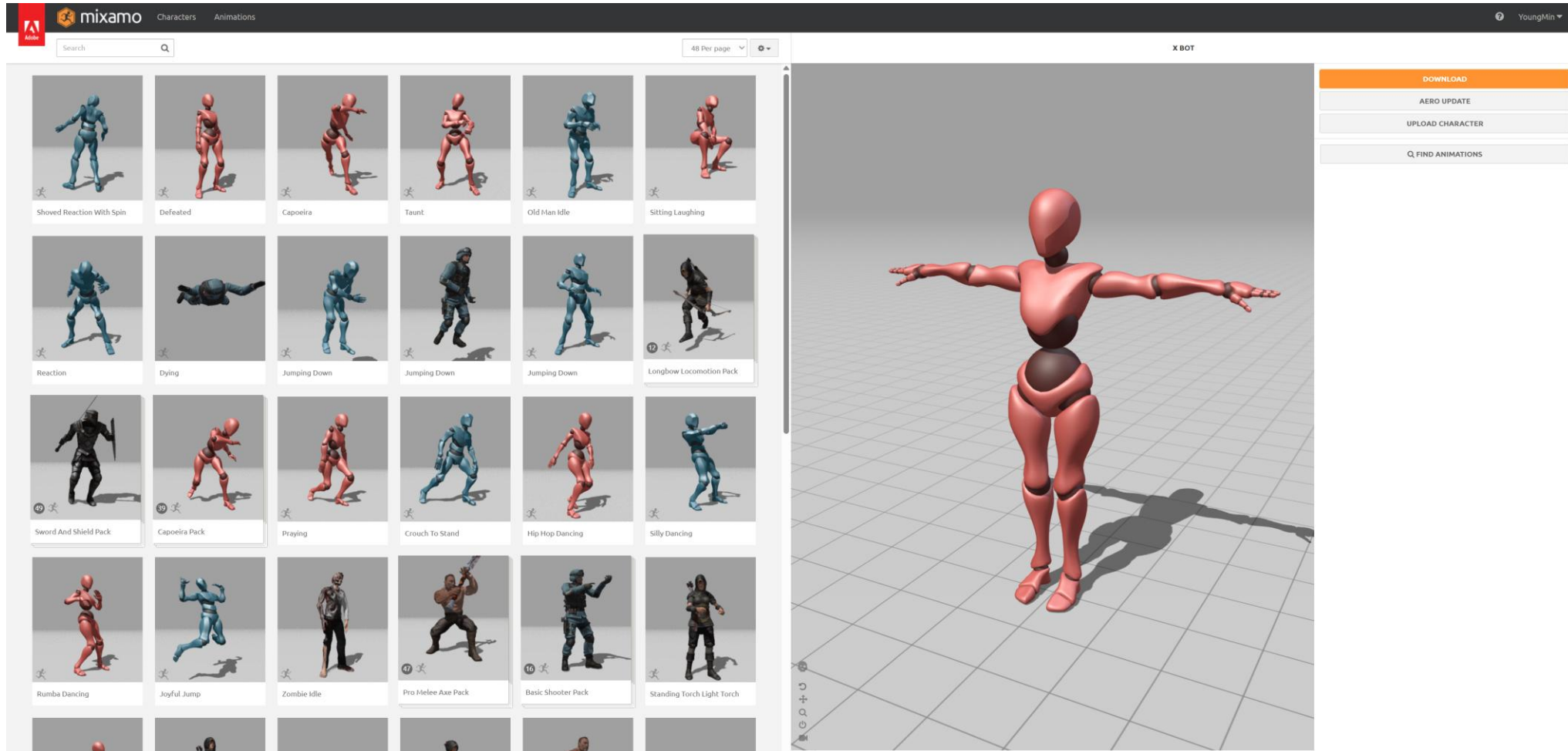
EUR

Blender: OpenSource DCC

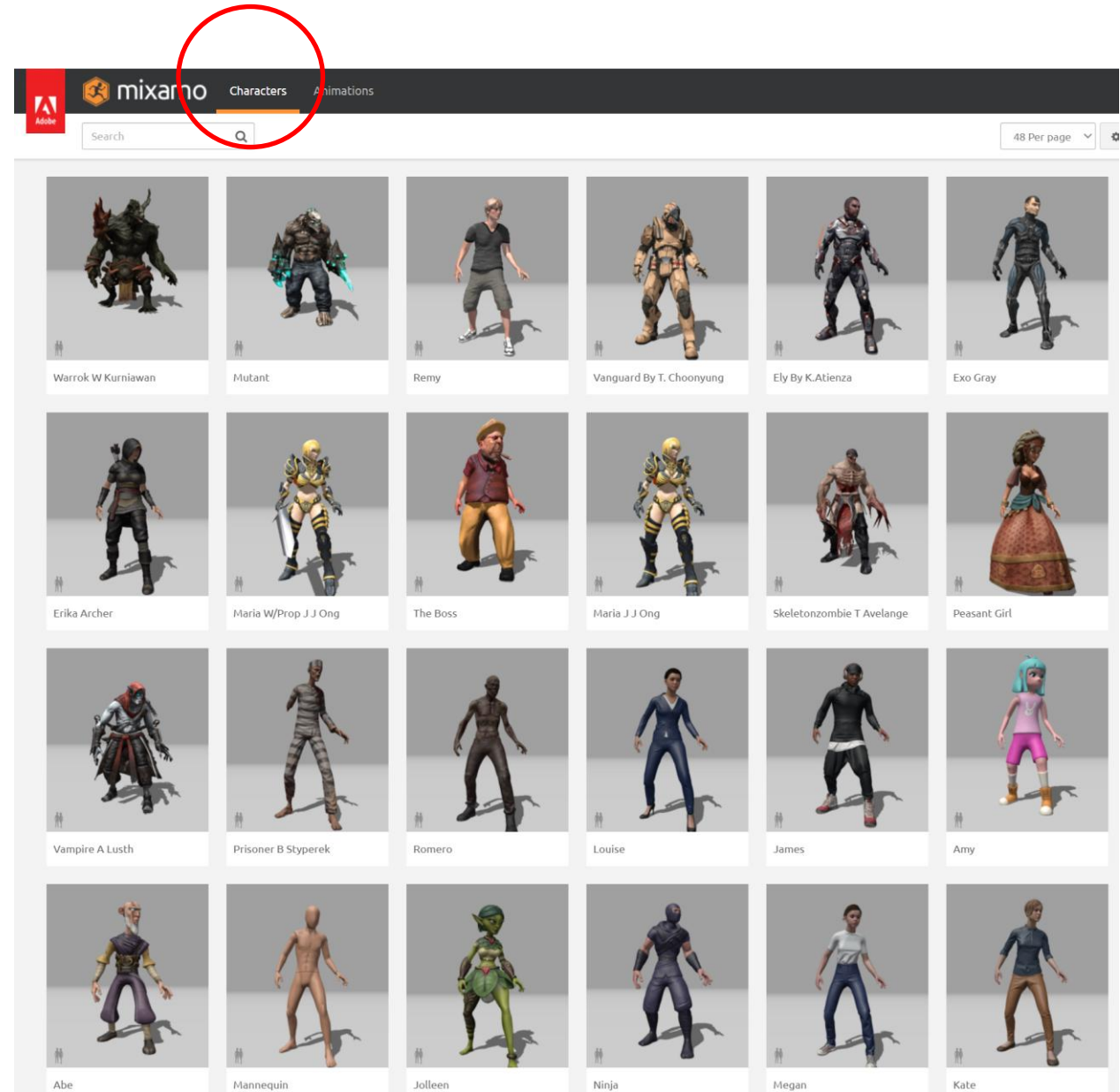


Mixamo에 접속해 보자

<https://www.mixamo.com/>

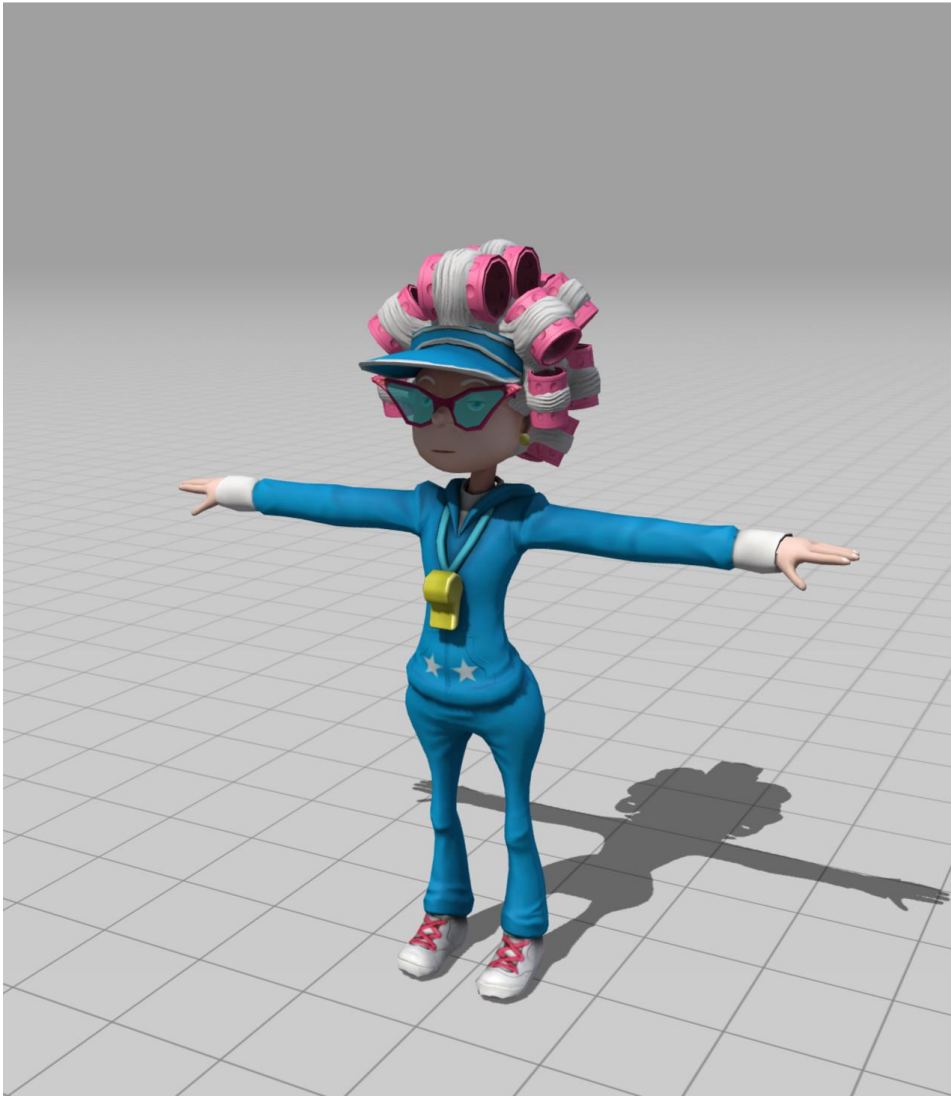


필요한 캐릭터를 선택해 보자



다운로드: sporty granny

SPORTY GRANNY



DOWNLOAD

AERO UPDATE

UPLOAD CHARACTER

Q, FIND ANIMATIONS

DOWNLOAD SETTINGS

Format

FBX for Unity(.fbx)

Pose

T-pose

CANCEL

DOWNLOAD

FBX for Unity 타입으로 다운로드

애니메이션 데이터 찾기

Adobe mixamo Characters Animations

idle

48 Per page

× "Idle"

Showing 1-48 of 507 results

Fight Idle To Standing Idle

Action Idle To Fight Idle

Action Idle To Standing Idle

Standing Idle To Fight Idle

Idle

Idle

Idle

Idle

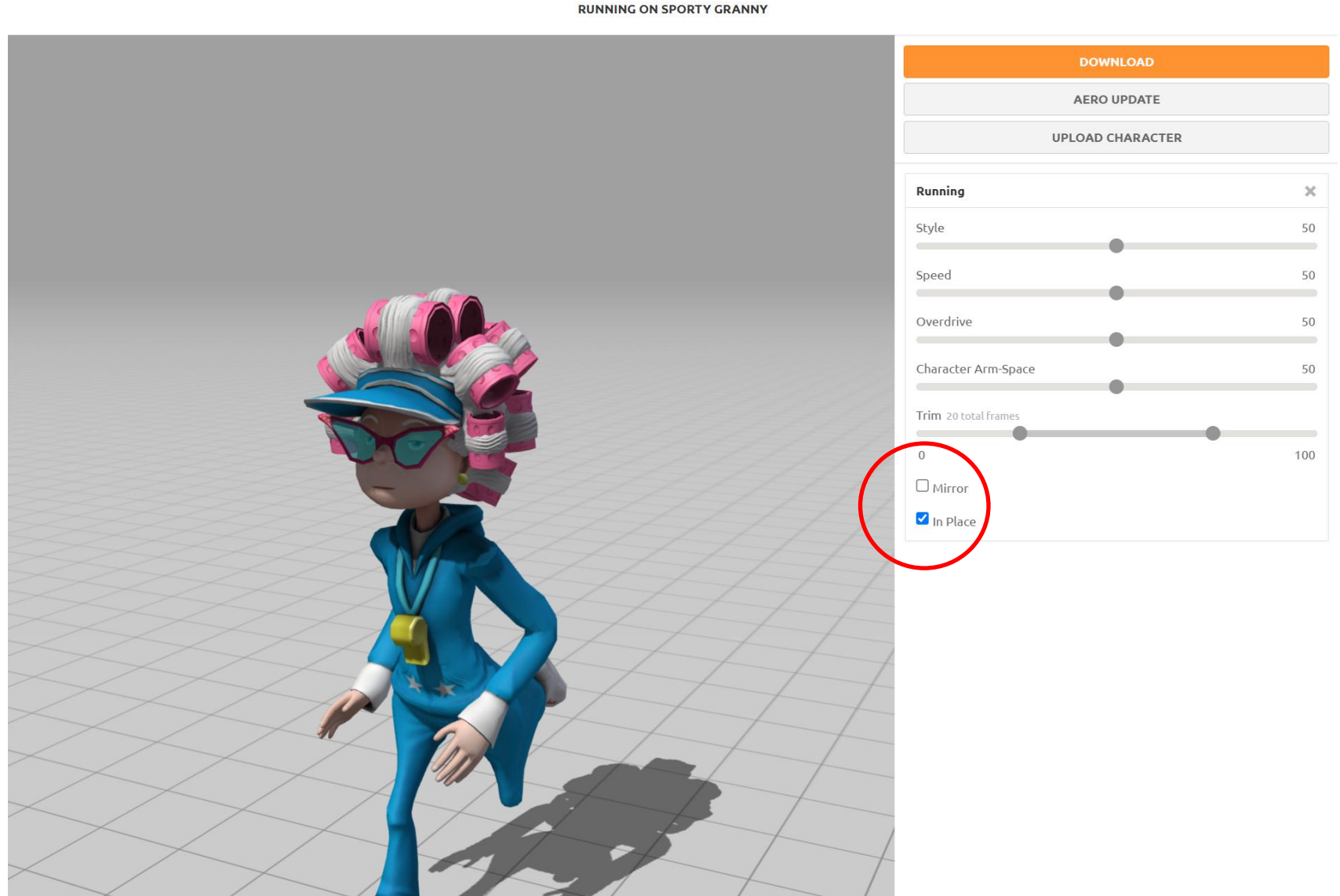
Kneeling Idle

Breathing Idle

Idle

Standing Idle

In Place Ani



다운로드

DOWNLOAD SETTINGS

Format

FBX for Unity(.fbx)



Skin

Without Skin



Frames per Second

60



Keyframe Reduction

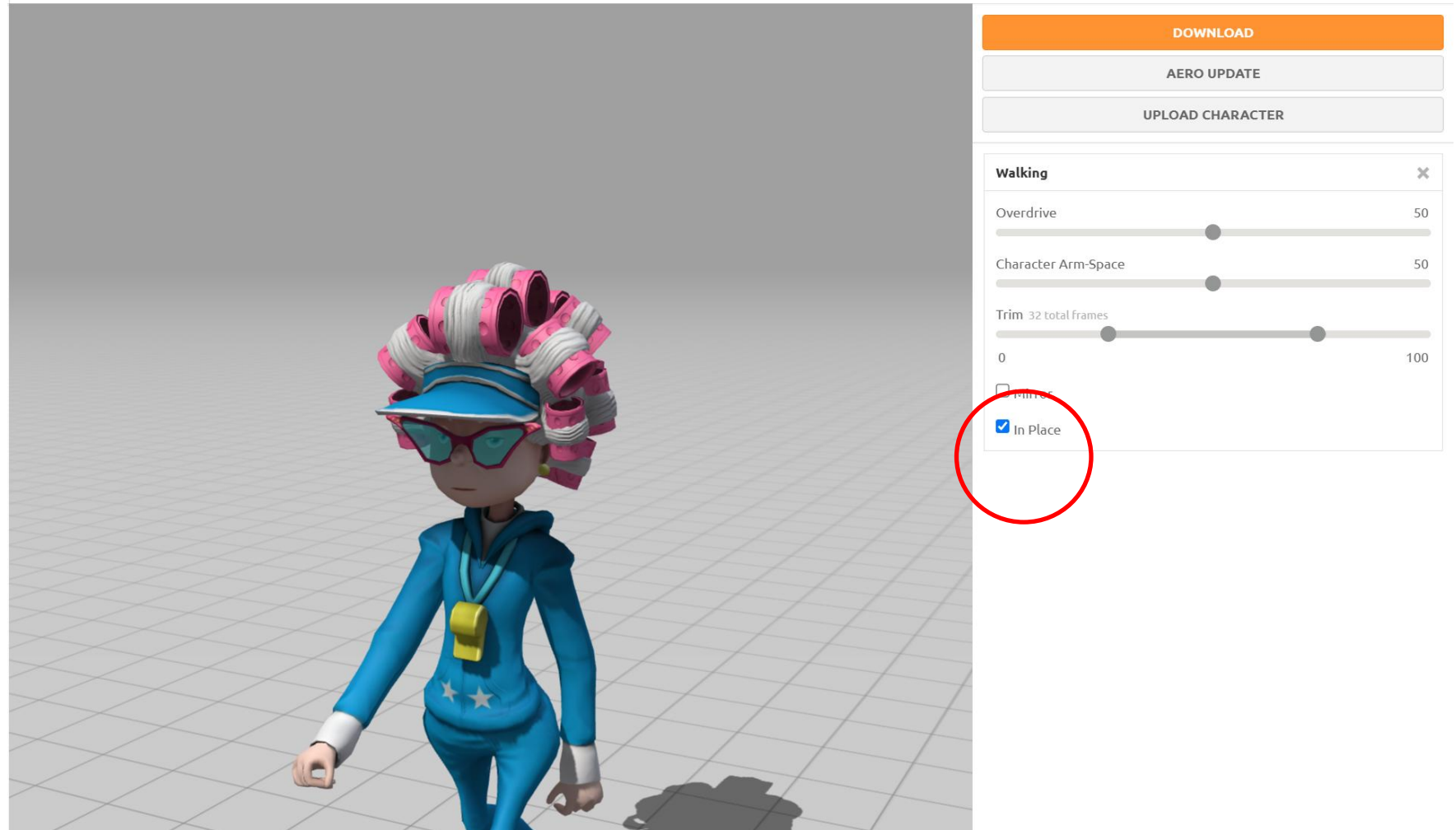
none



CANCEL

DOWNLOAD

다양한 애니메이션 다운로드

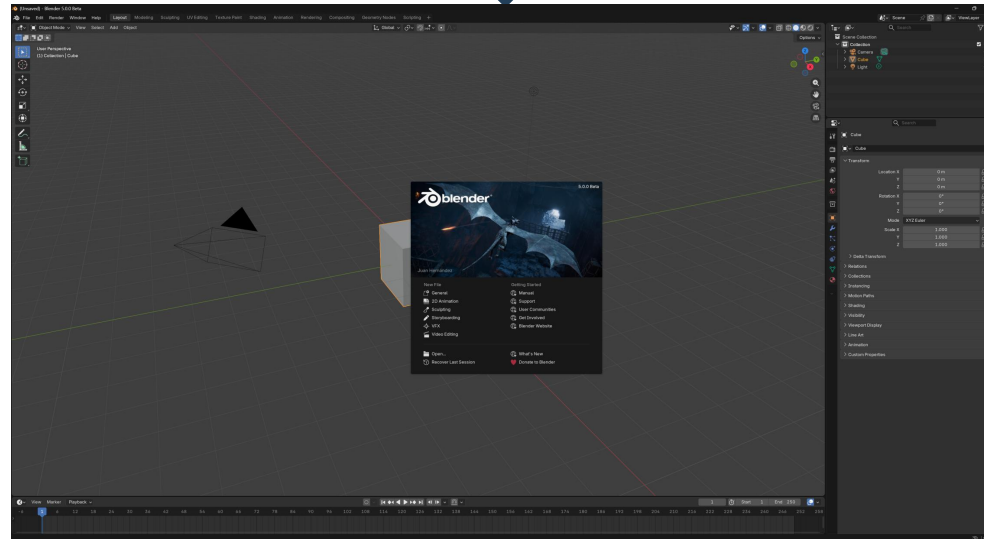


하나로 합치자

 Sporty Granny@Walking	11/6/2025 9:55 AM	3D Object
 Sporty Granny@Running	11/6/2025 9:54 AM	3D Object
 Sporty Granny@Idle	11/6/2025 9:53 AM	3D Object
 Sporty Granny	11/6/2025 9:51 AM	3D Object



블렌더
Blender



다른 방법: Character animation combiner

<https://nilooy.github.io/character-animation-combiner/>



캐릭터와 애니메이션 설정

Character Animation Combiner

Upload File with Character

(.fbx / .gltf / .gltf)

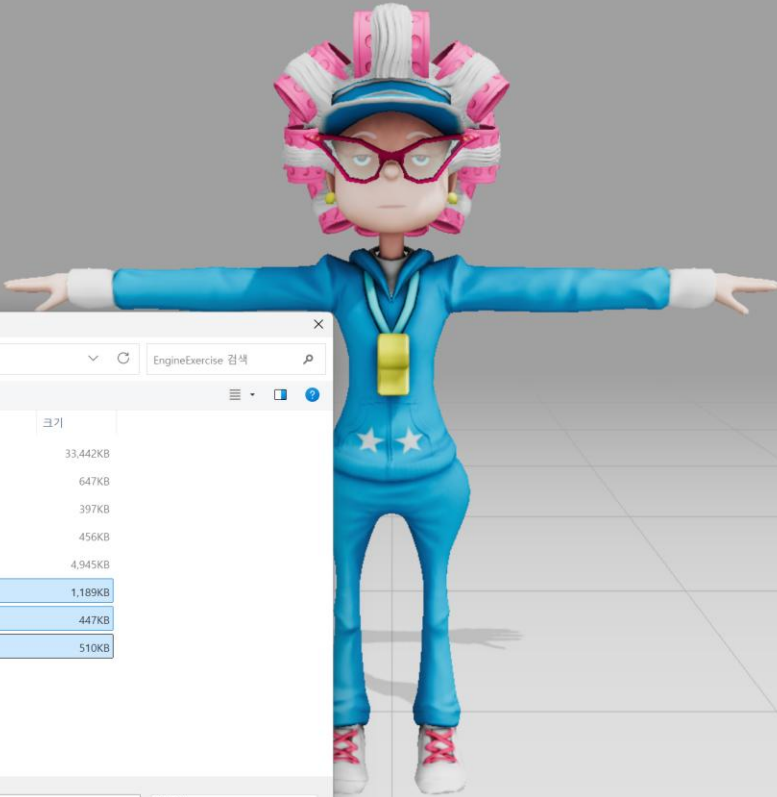
UPLOAD FILE Sporty Granny.fbx

Upload Animations (.fbx)

UPLOAD FILE

Export

EXPORT GLTFEXPORT GLB



Animations (2)

Double Click to rename | Click to play

Stop All Animations

T-Pose (No Animation)

mixamo.com

열기

문서 > YMKang_Work > 수업_실습 > 2025_2학기 > 2025_2_게임연진기초 > 강의노트 > EngineExercise

EngineExercise 검색

구성 새 폴더

문서

배경 화면

사진

첨부 파일

배경 화면

다운로드

문서

사진

음악

동영상

04_수상작

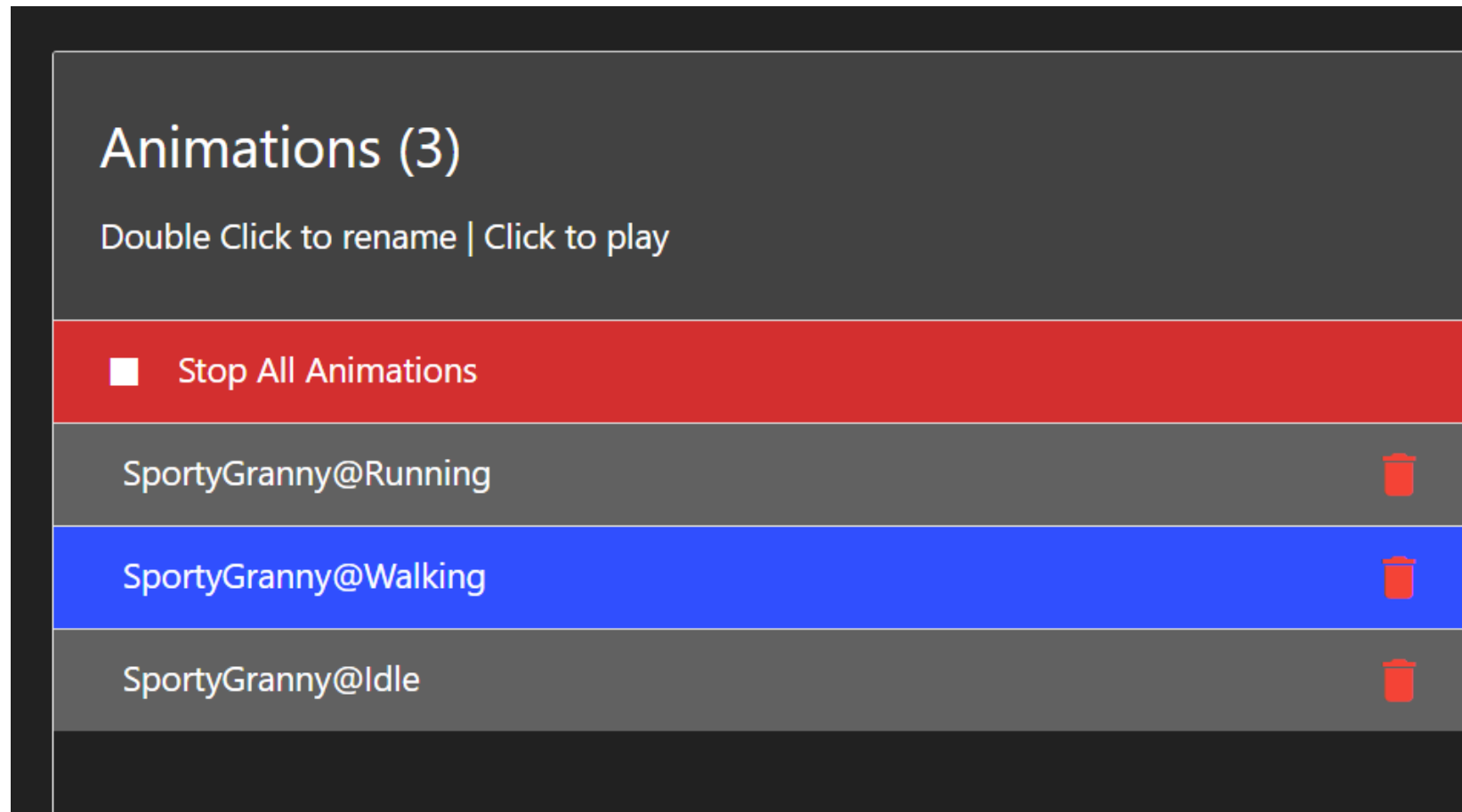
스크린샷

이름	상태	수정된 날짜	유형	크기
Ch09_nonPBR	✓	11/5/2025 12:32 PM	3D Object	33,442KB
Ch09_nonPBR@Idle	✓	11/5/2025 12:38 PM	3D Object	647KB
Ch09_nonPBR@Running	✓	11/5/2025 1:37 PM	3D Object	397KB
Ch09_nonPBR@Walking	✓	11/5/2025 1:38 PM	3D Object	456KB
Sporty Granny	✓	11/6/2025 9:51 AM	3D Object	4,945KB
Sporty Granny@Idle	✓	11/6/2025 9:53 AM	3D Object	1,189KB
Sporty Granny@Running	✓	11/6/2025 9:54 AM	3D Object	447KB
Sporty Granny@Walking	✓	11/6/2025 9:55 AM	3D Object	510KB

파일 이름(N): "Sporty Granny@Walking" "Sporty Granny@Idle" "Sporty Granny@Running"FBX File

열기(O)취소

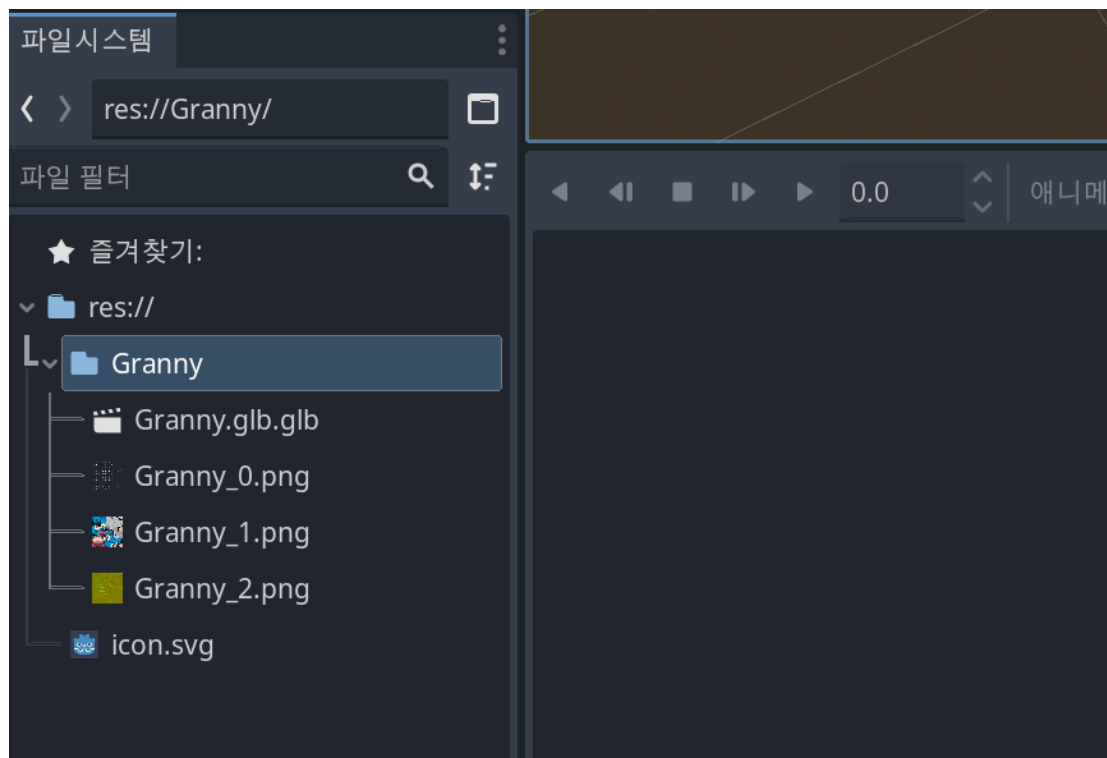
T pose 등 필요 없는 애니메이션 삭제



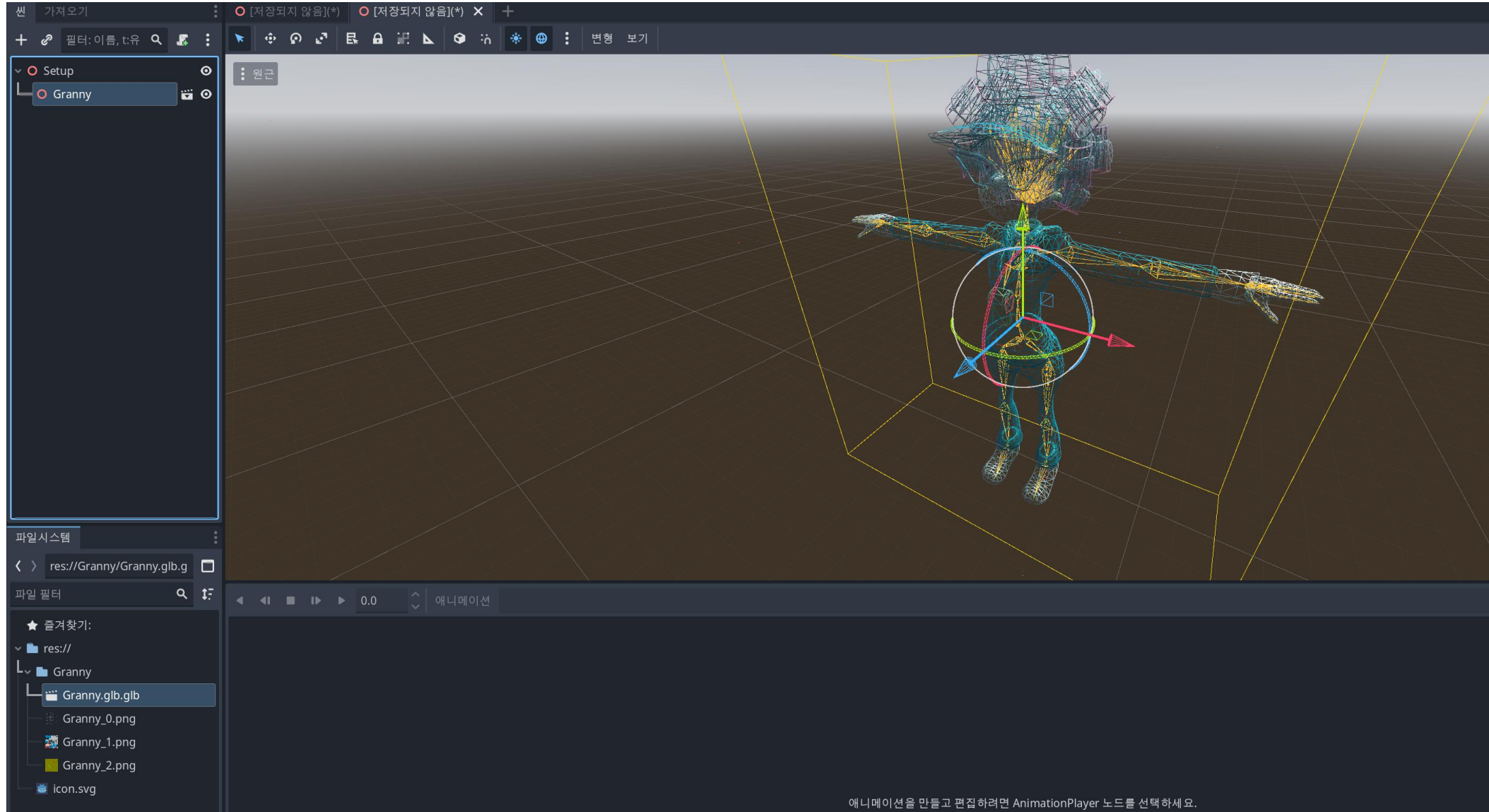
Export : glb

 Granny.glb		11/6/2025 10:59 AM	3D Object	13,805KB
--	---	--------------------	-----------	----------

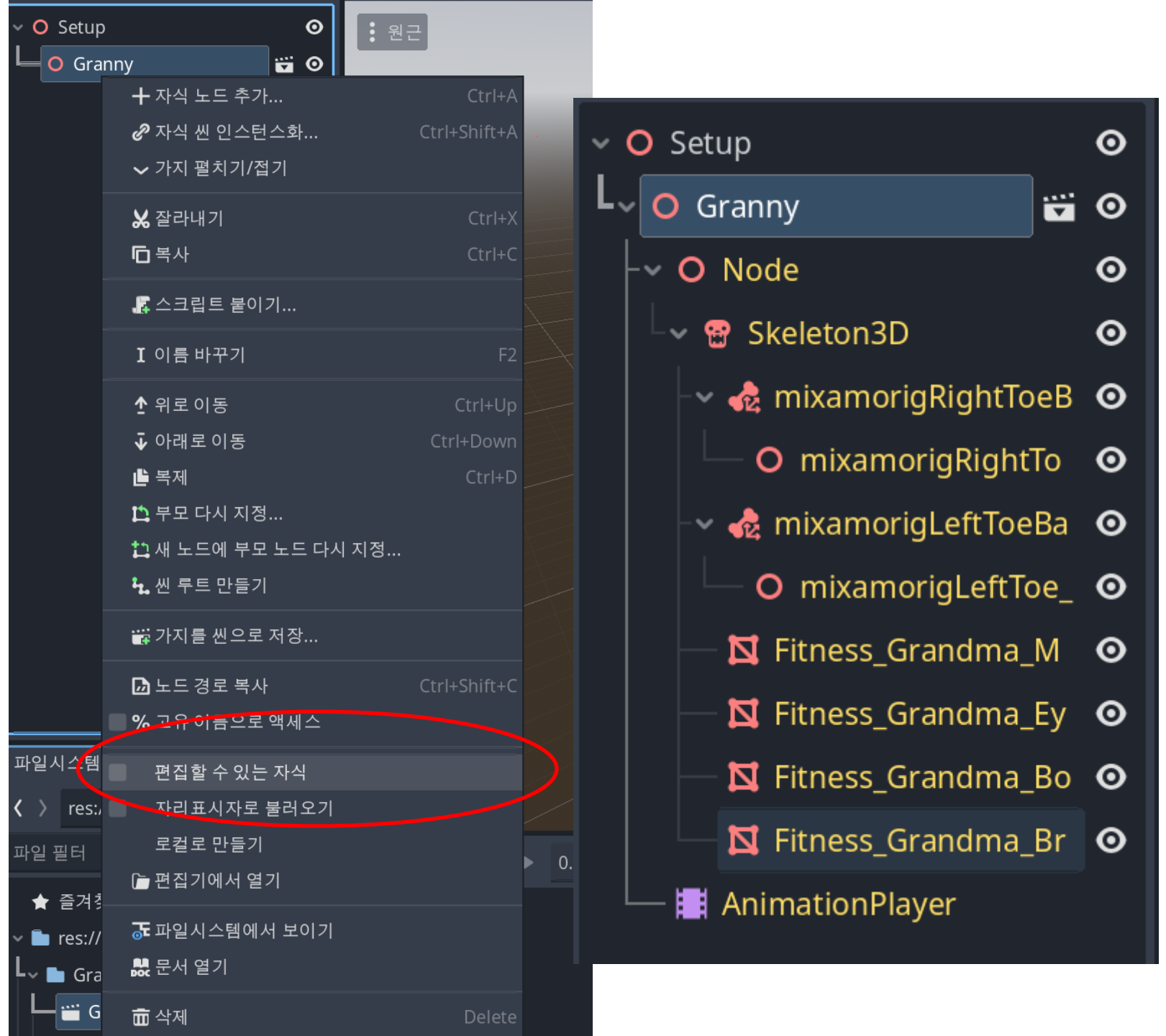
Godot 파일 시스템으로 옮기기



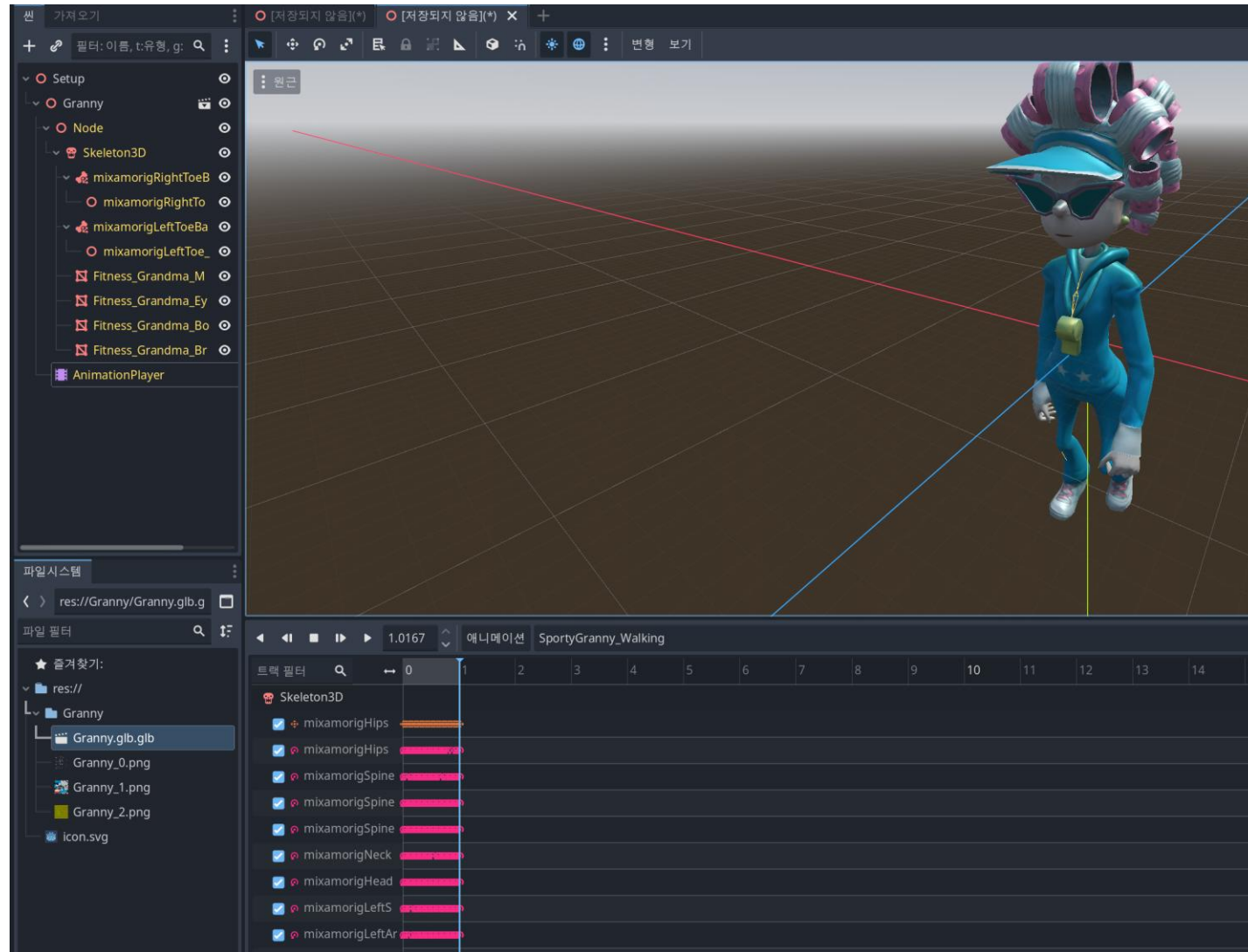
Scene



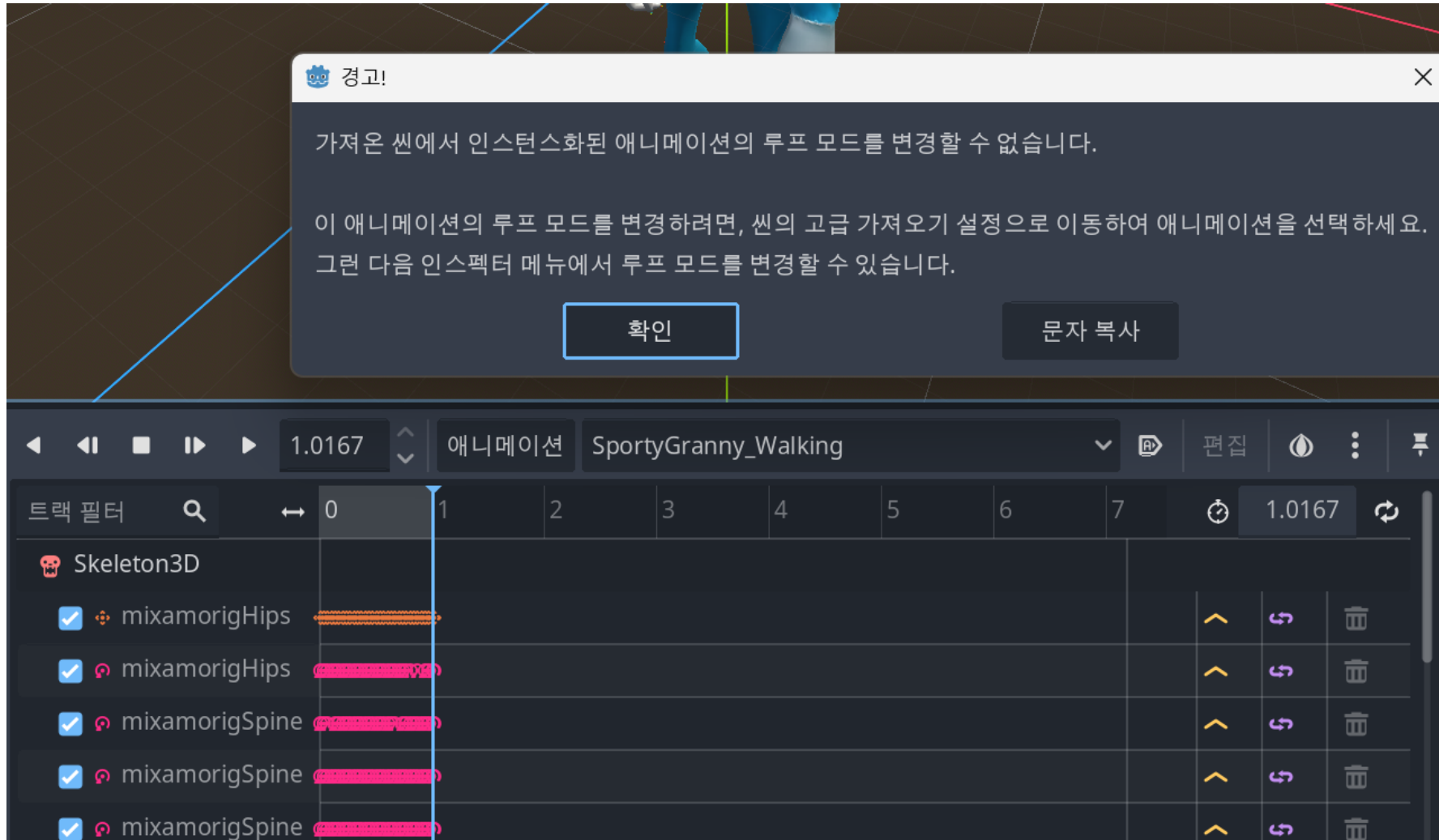
내부에 접근



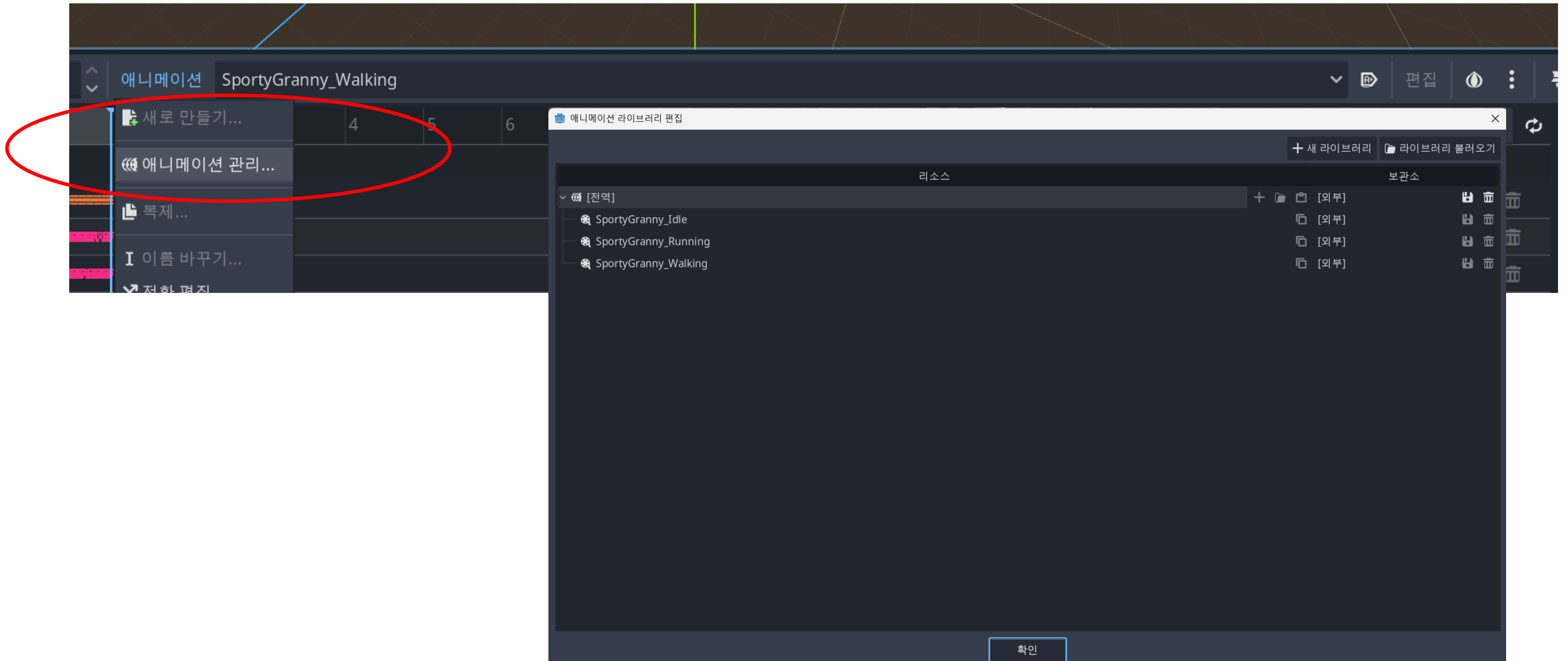
Animation Player



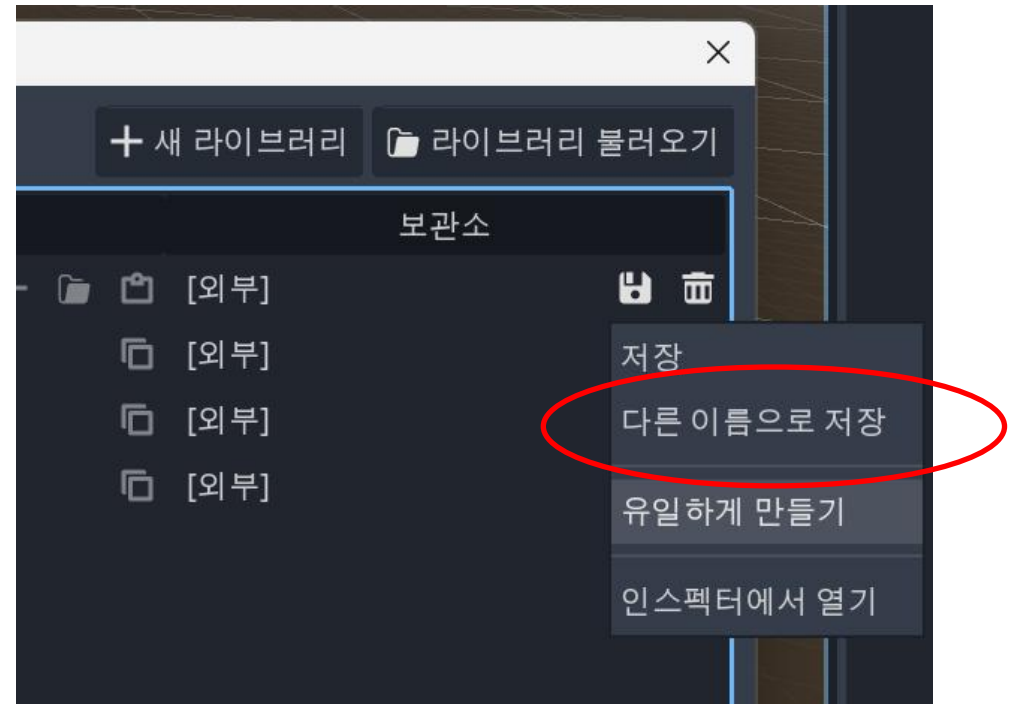
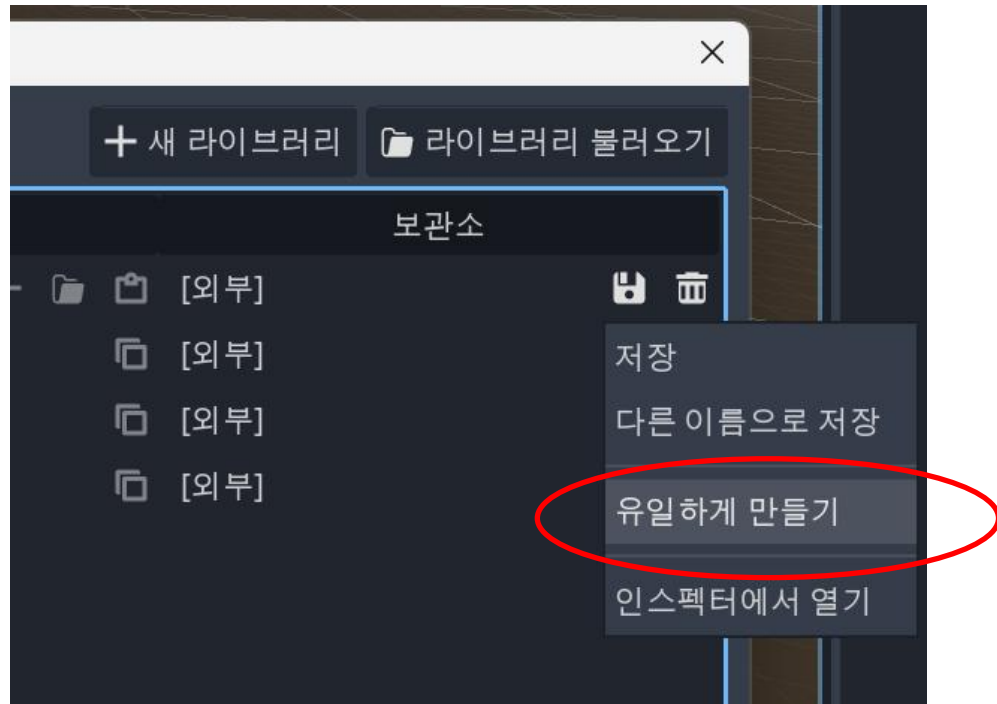
Animation Looping이 안 됨



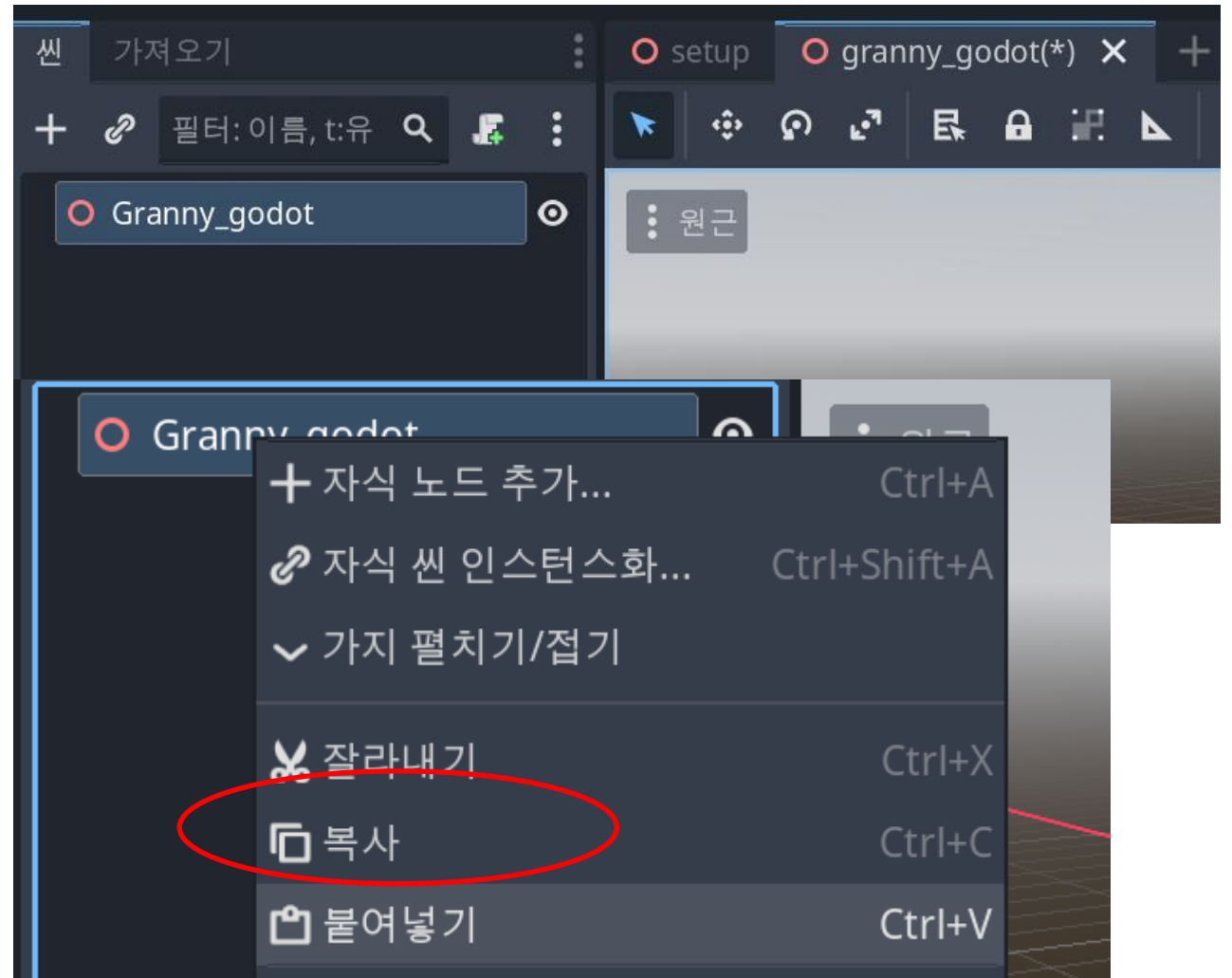
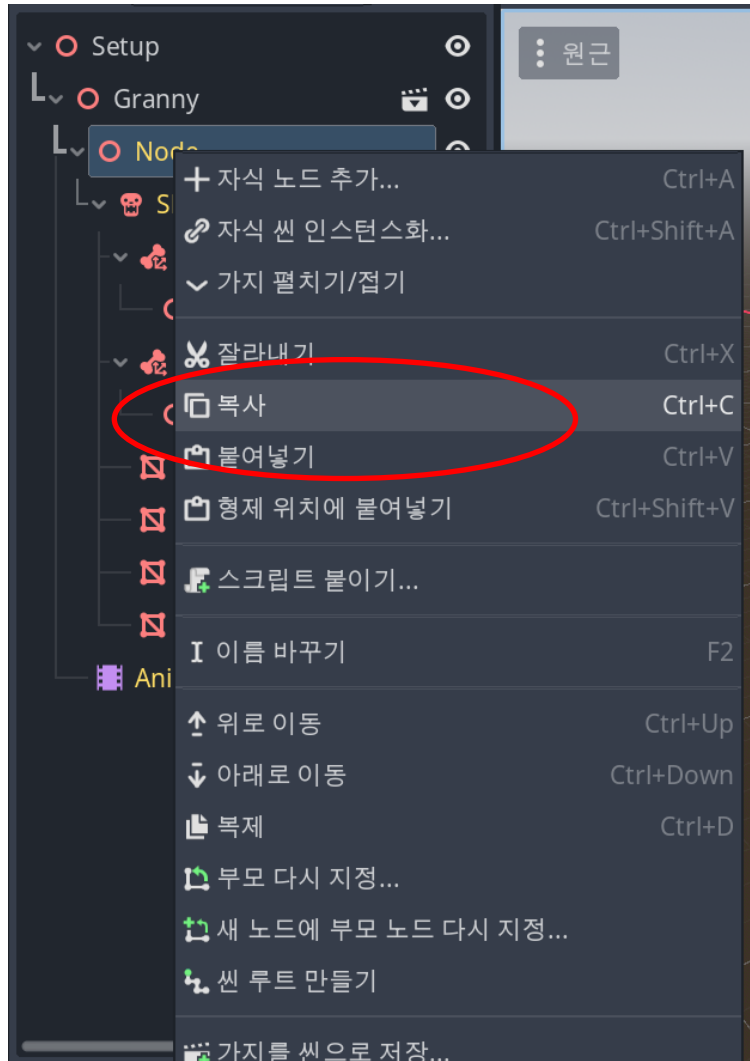
Animation 관리



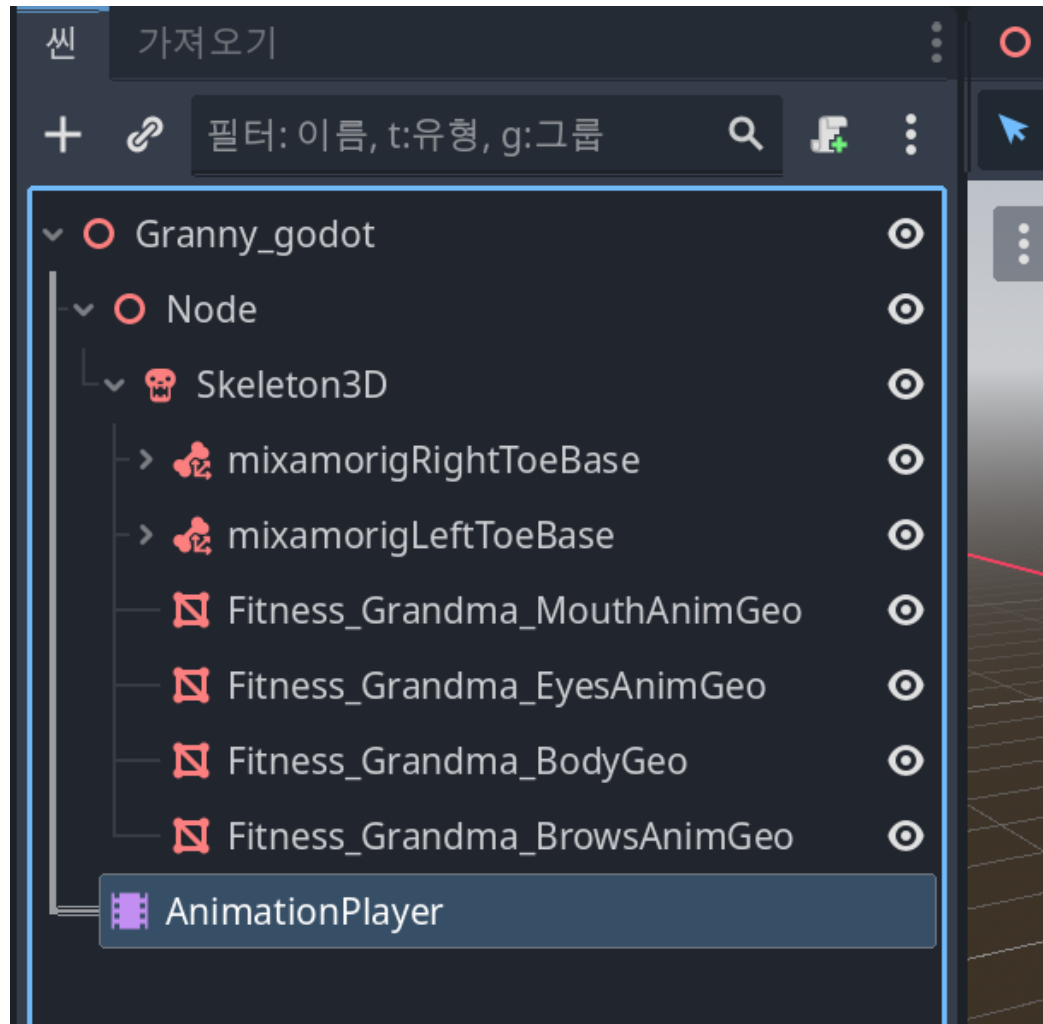
Animation 관리 ➔ 별도 저장



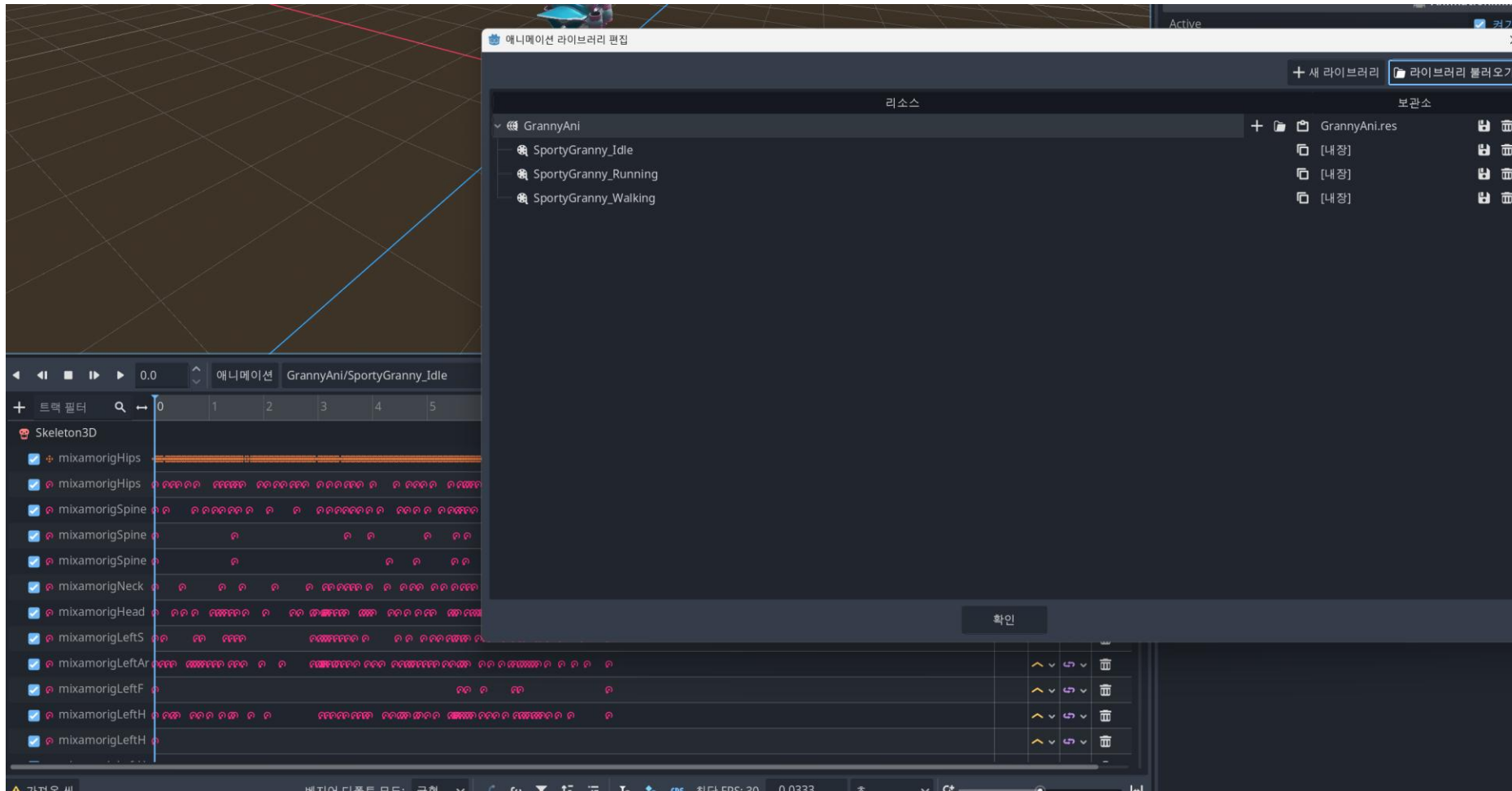
Armature 복사하기



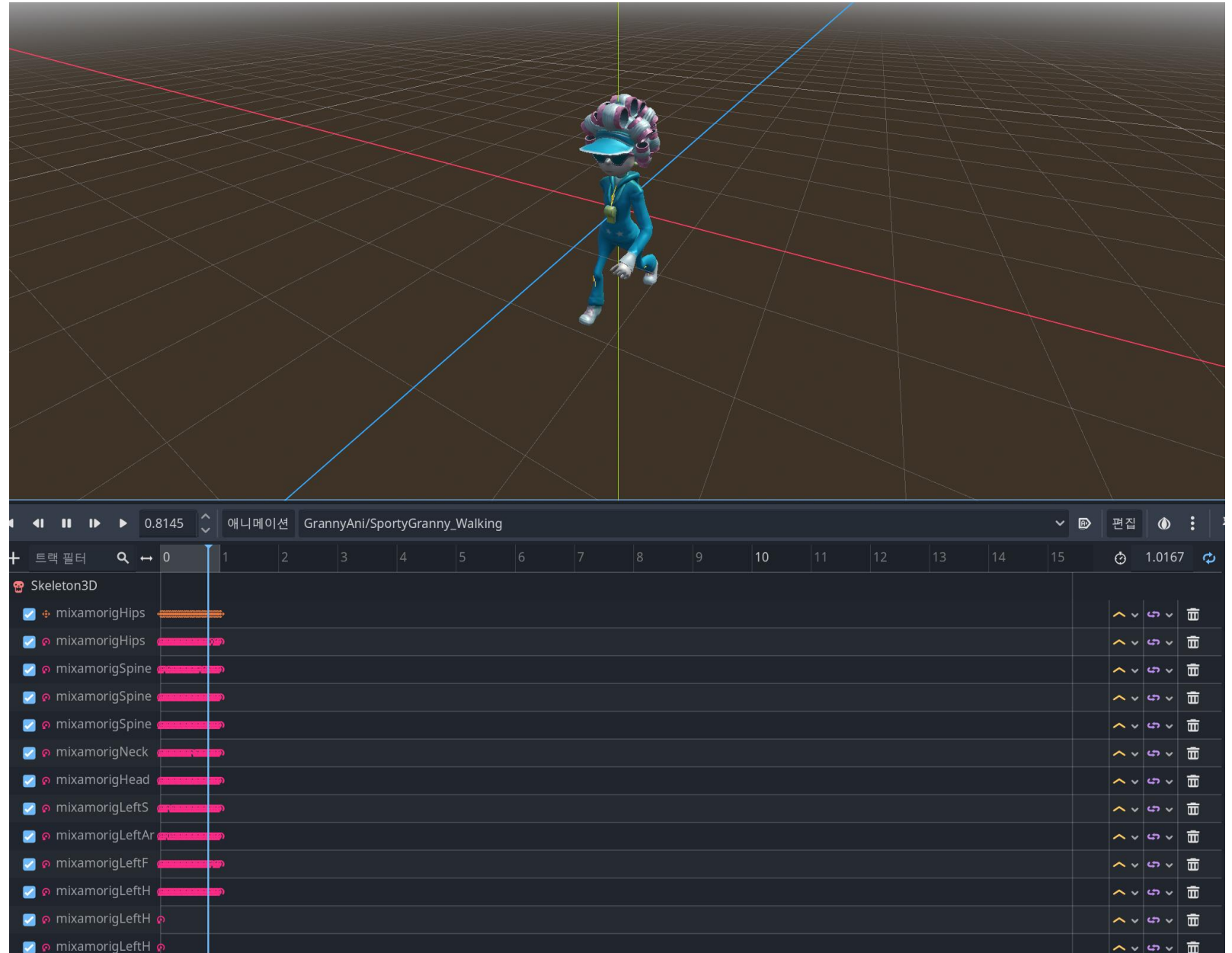
Animation Player 추가



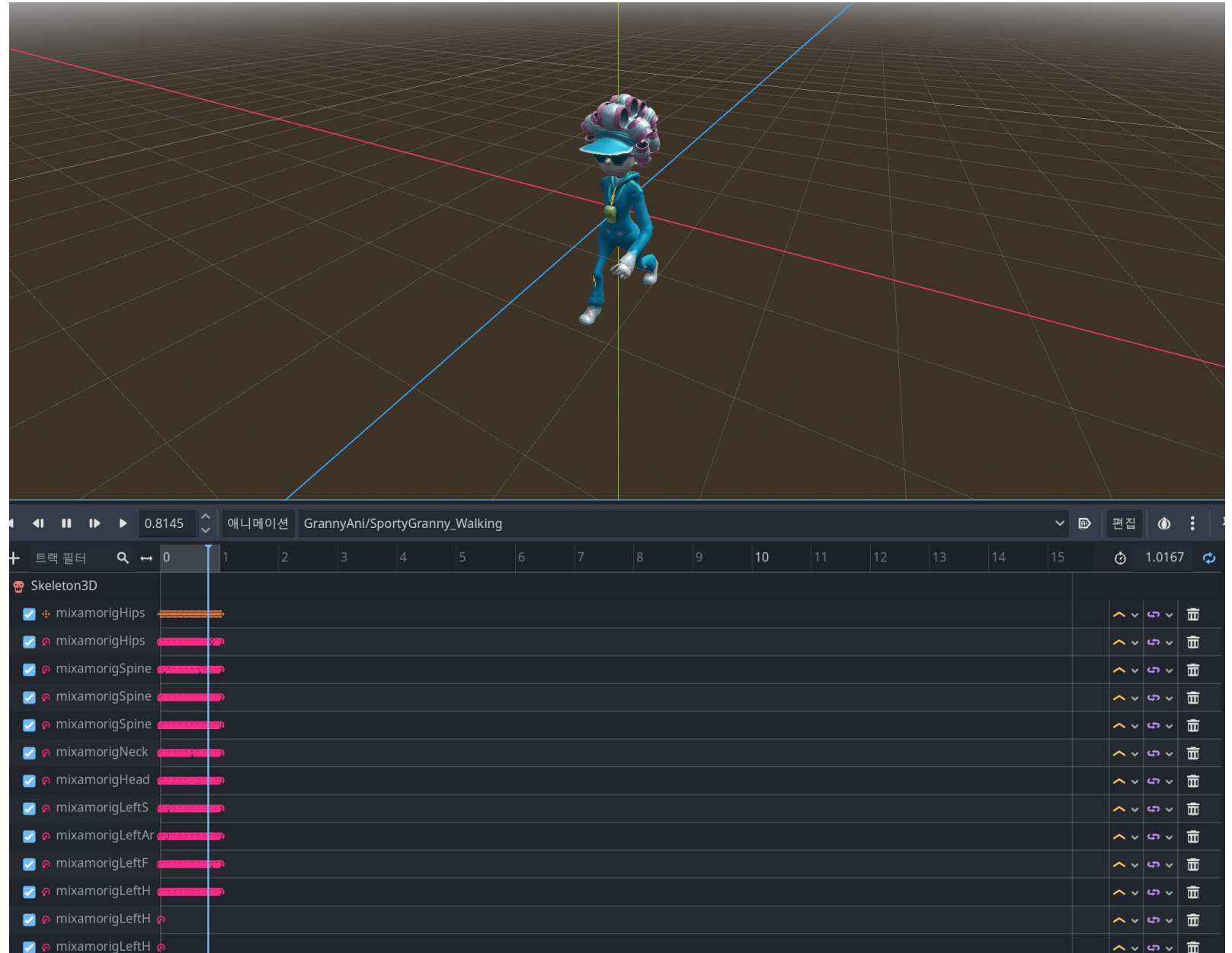
Animation 관리 → Load Library: GrannyAni.res



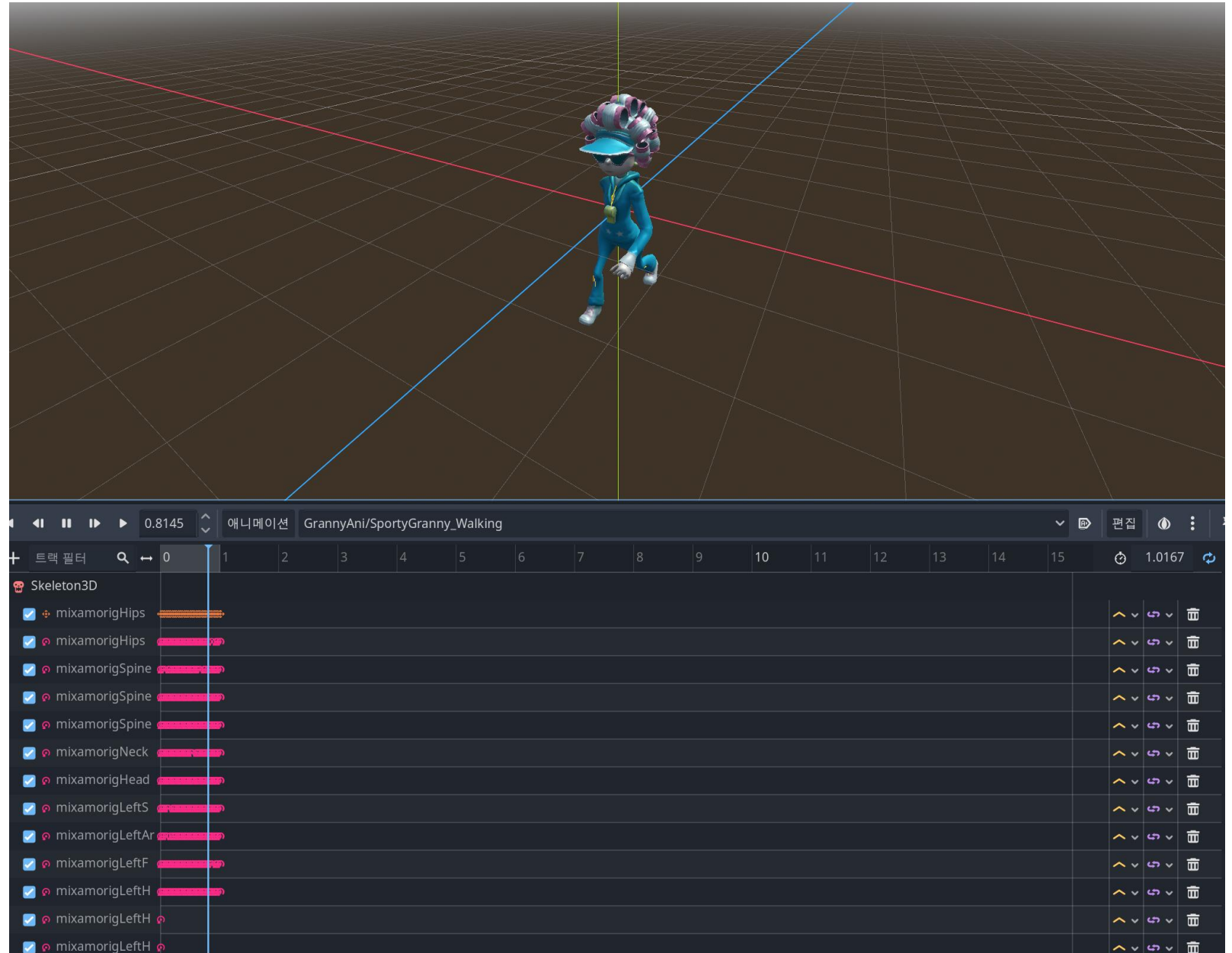
Looping OK



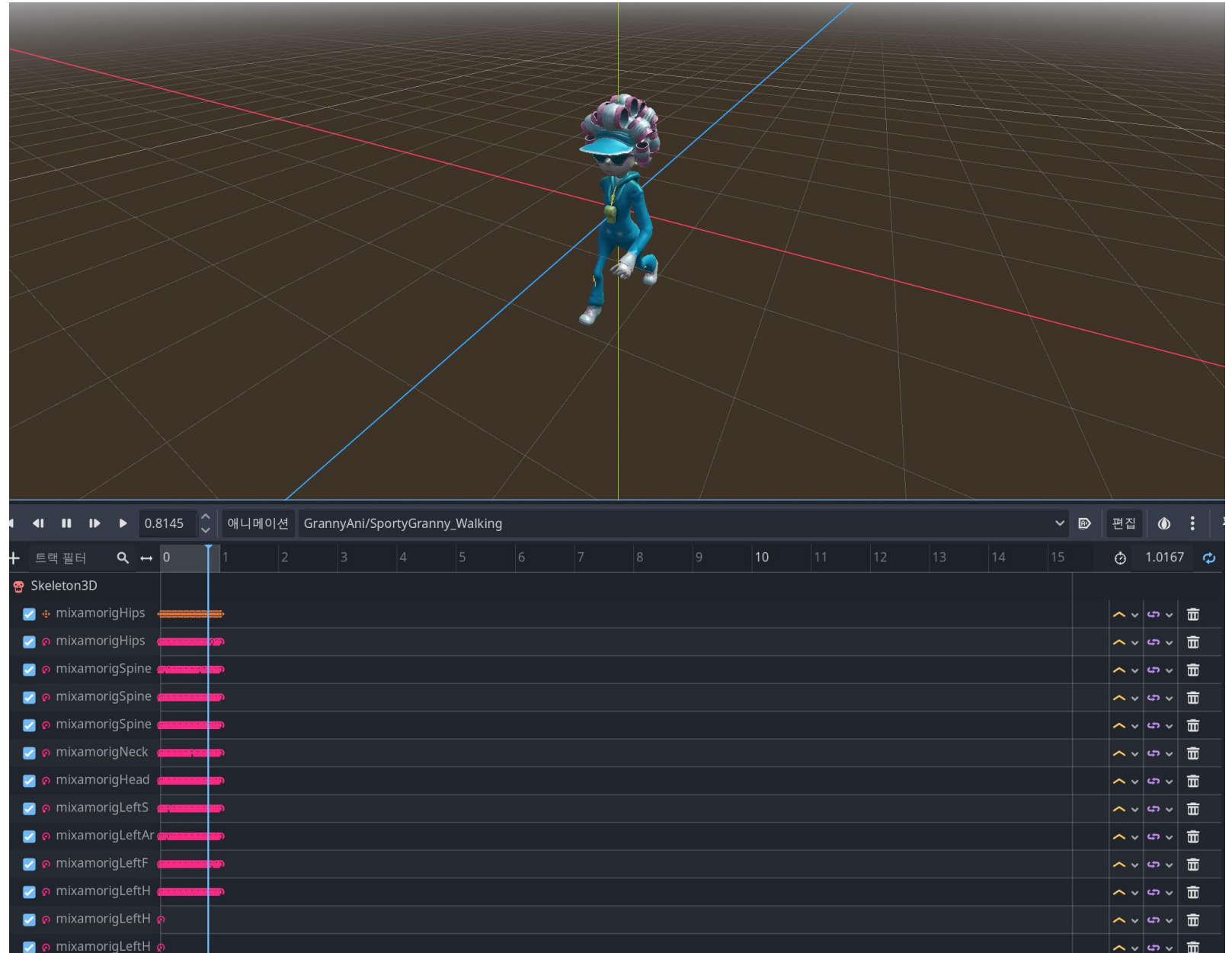
Looping OK



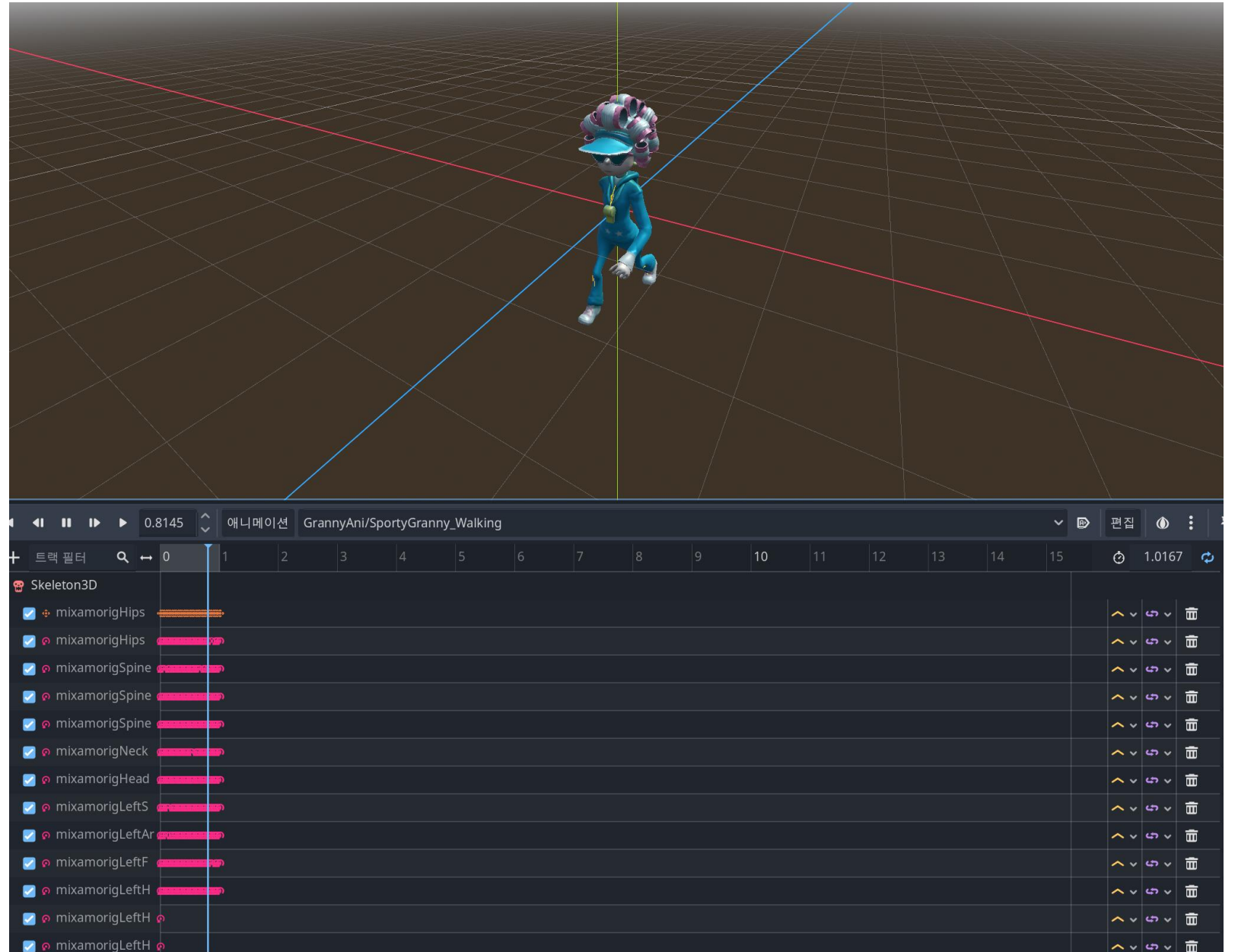
Looping OK



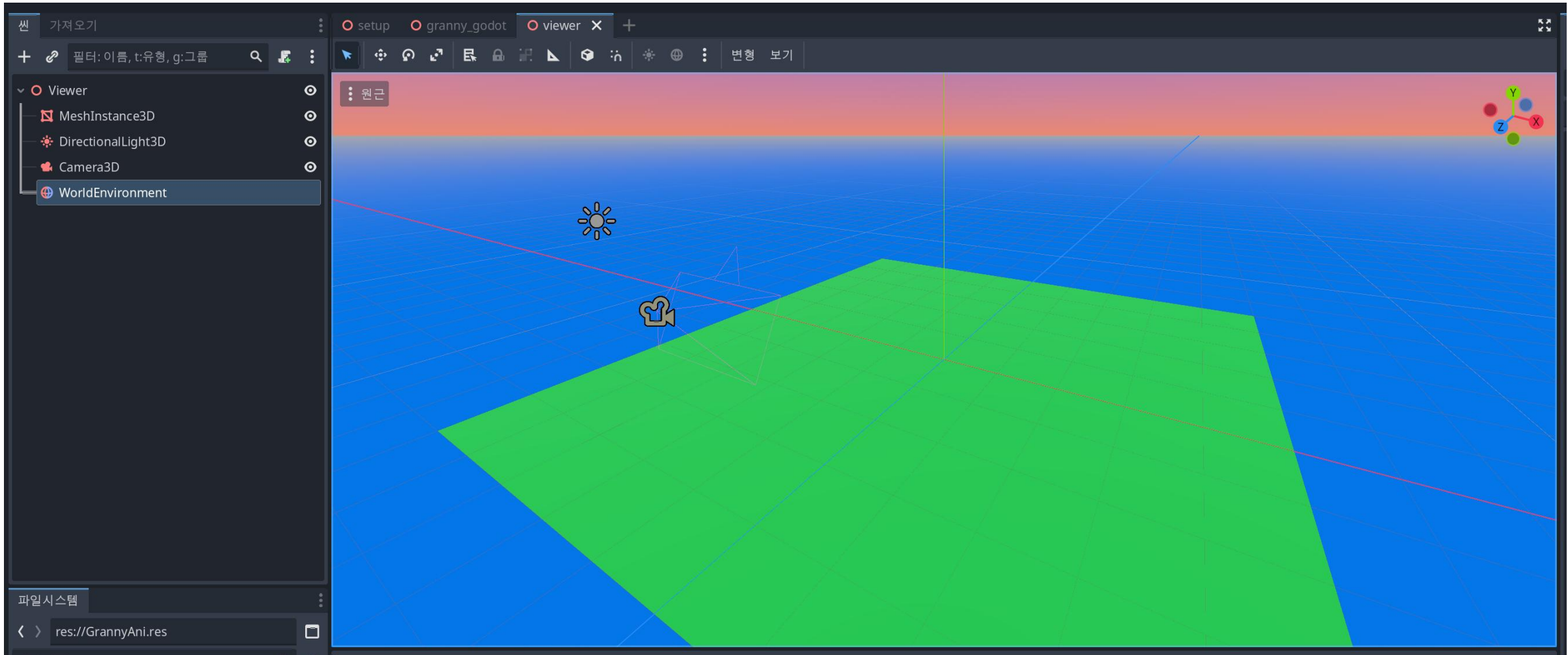
Looping OK



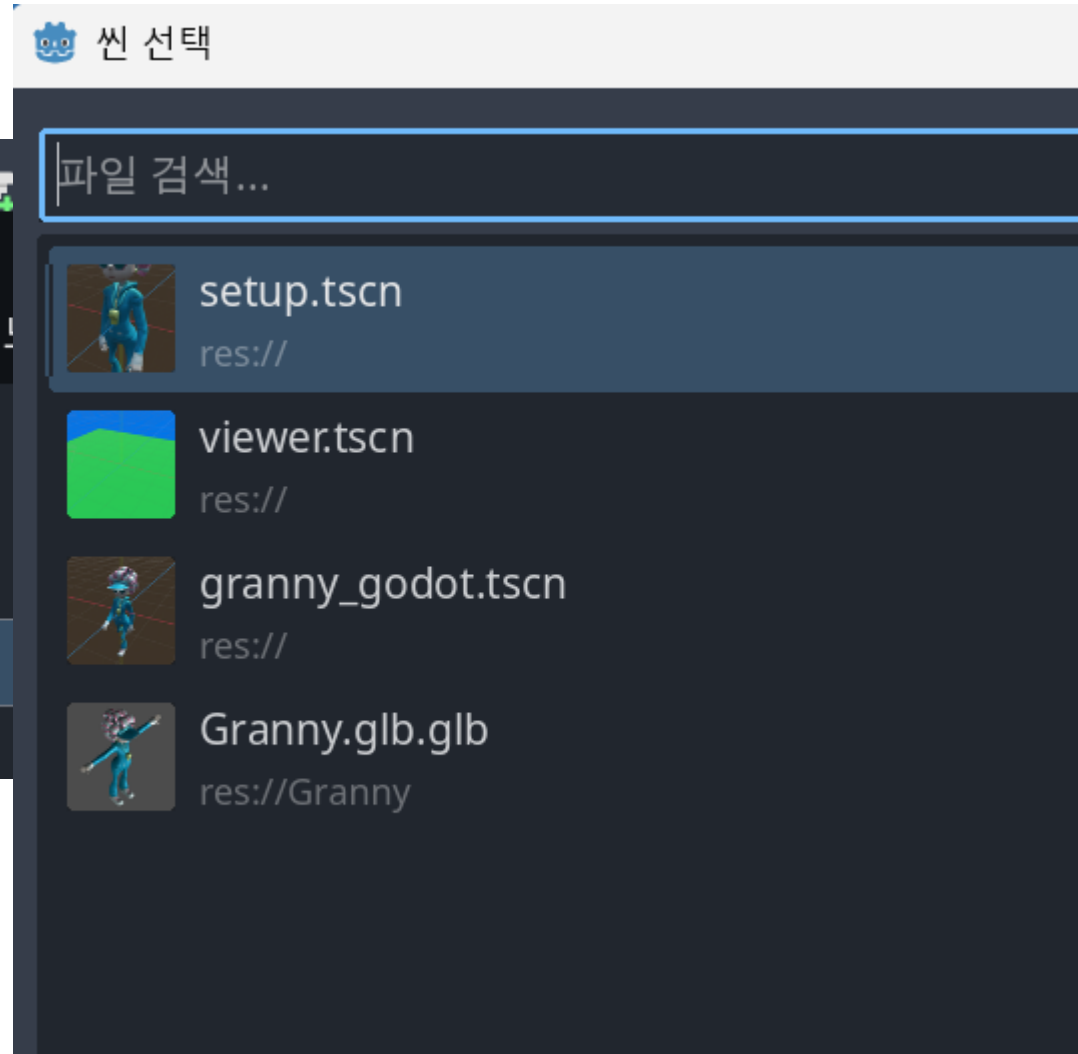
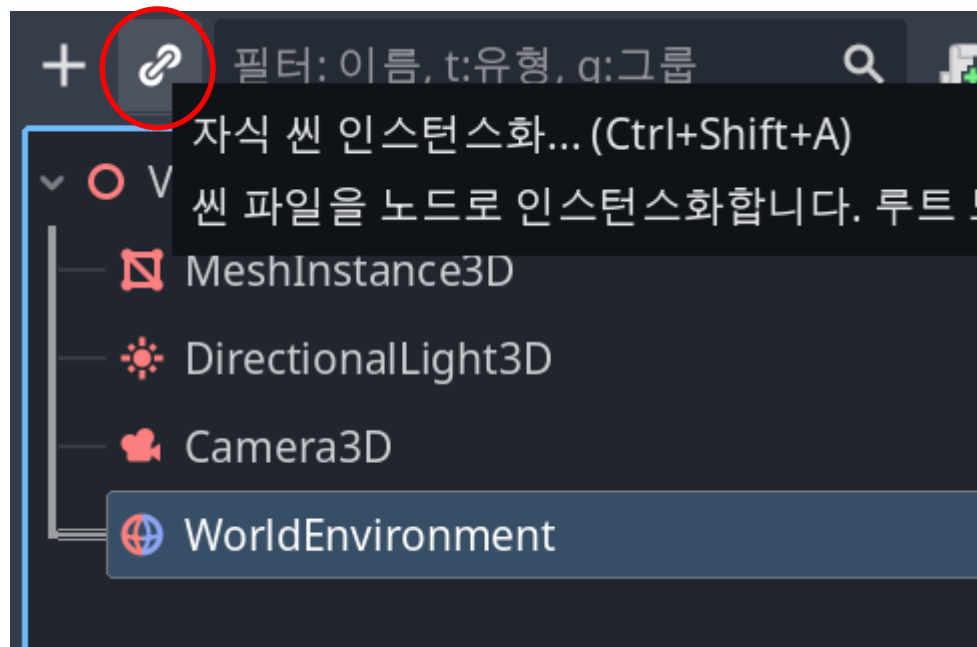
Looping OK



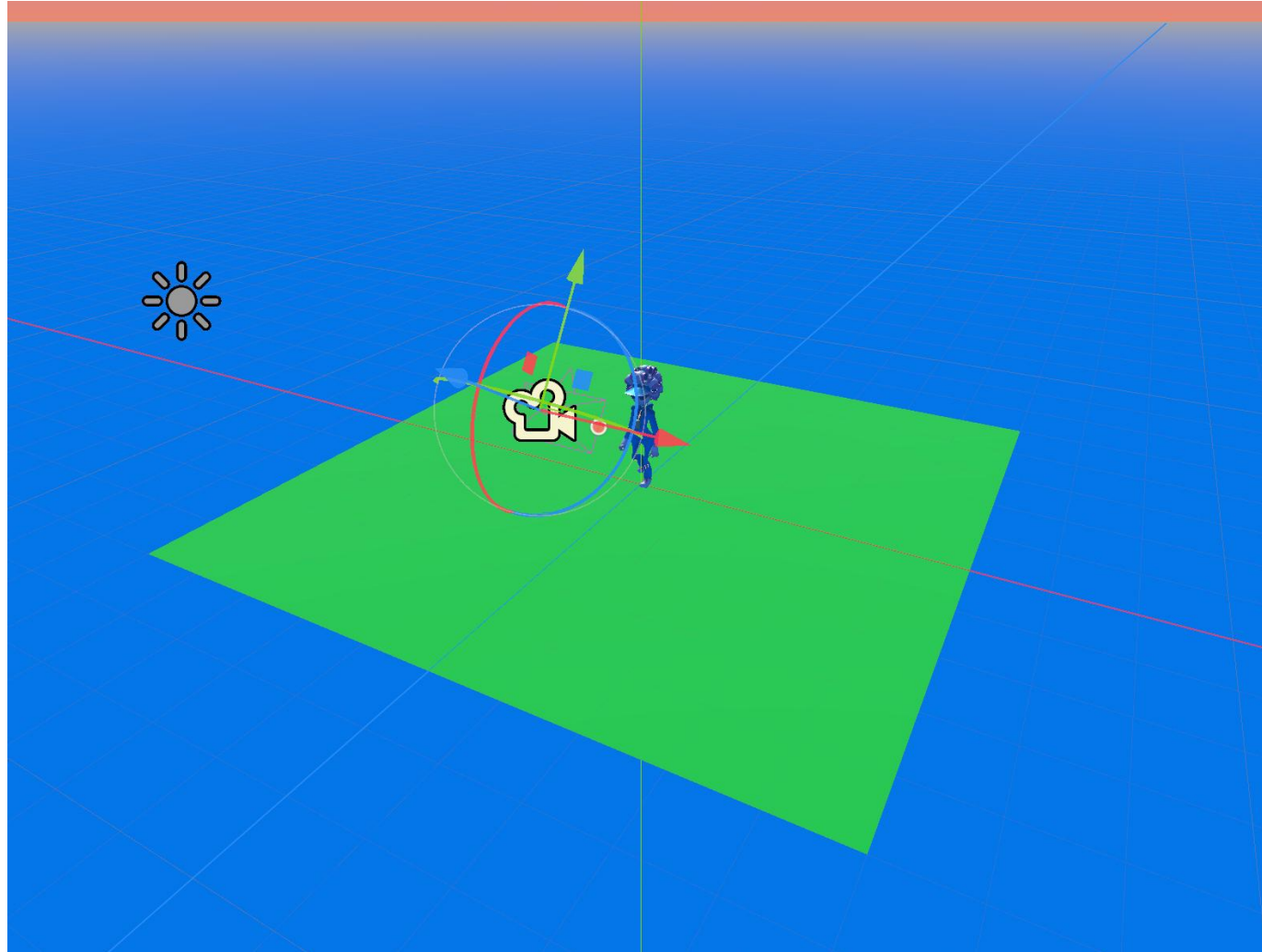
Viewer Scene 만들기



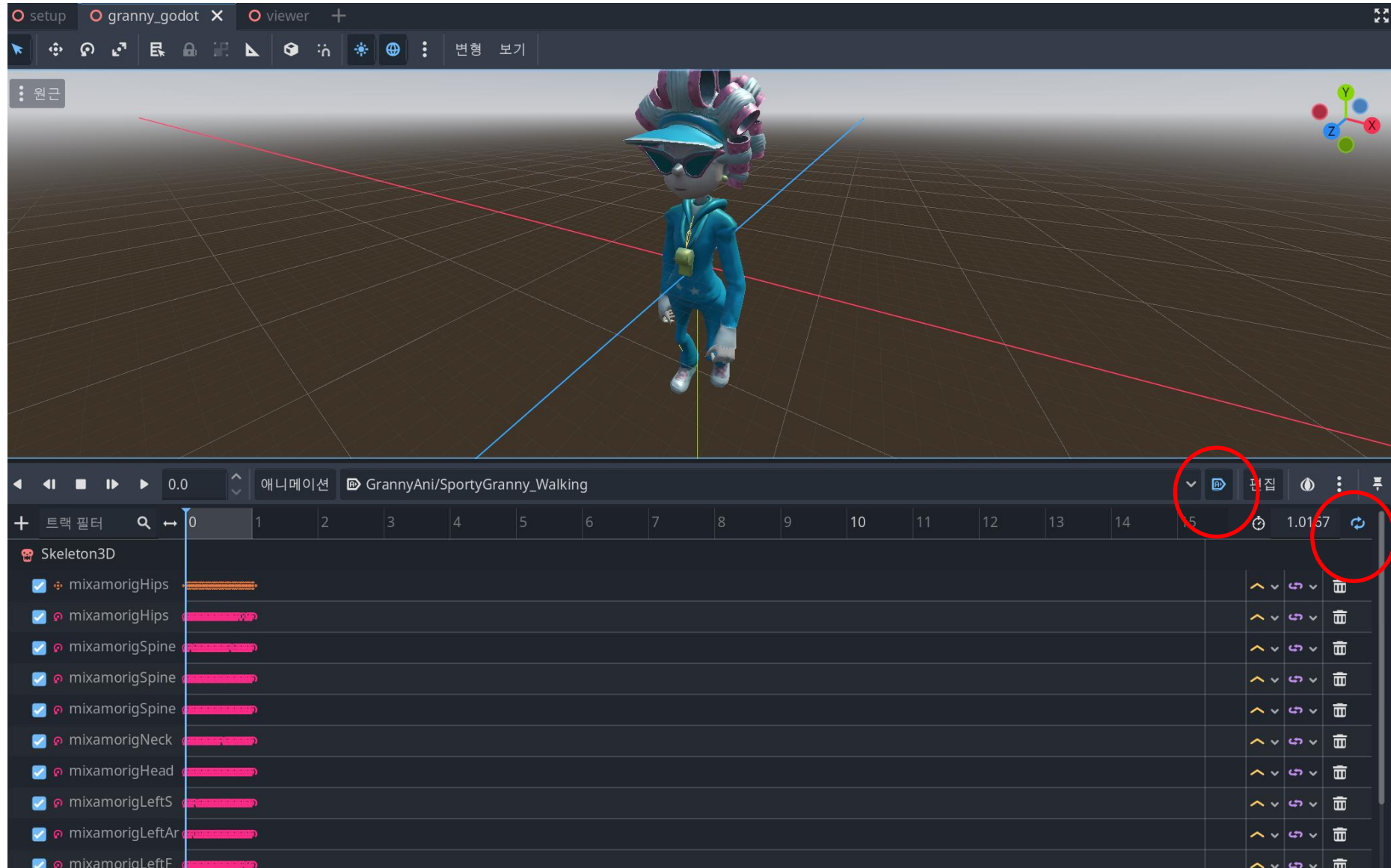
자식 씬 인스턴스화



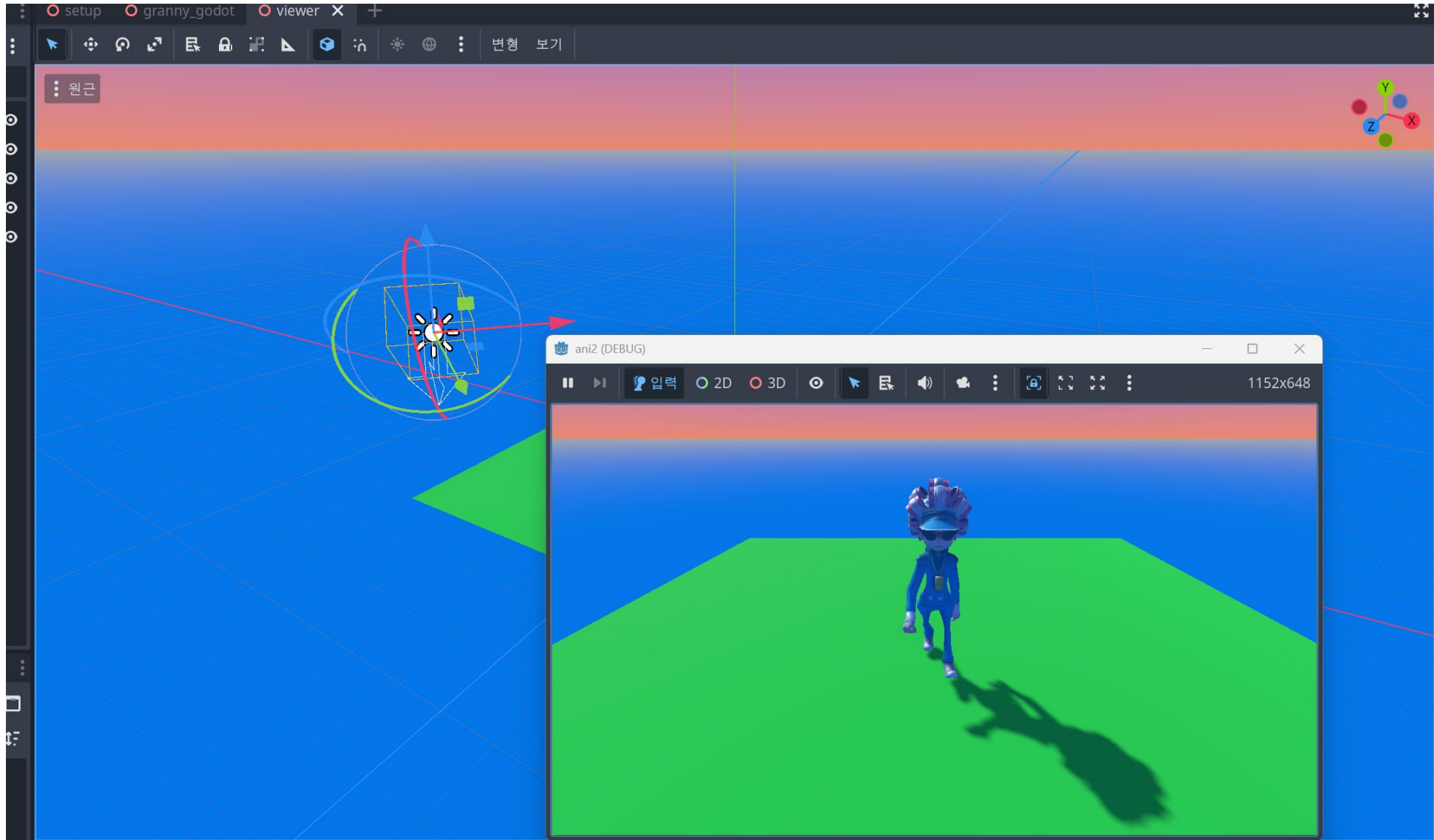
Viewer Scene 만들기



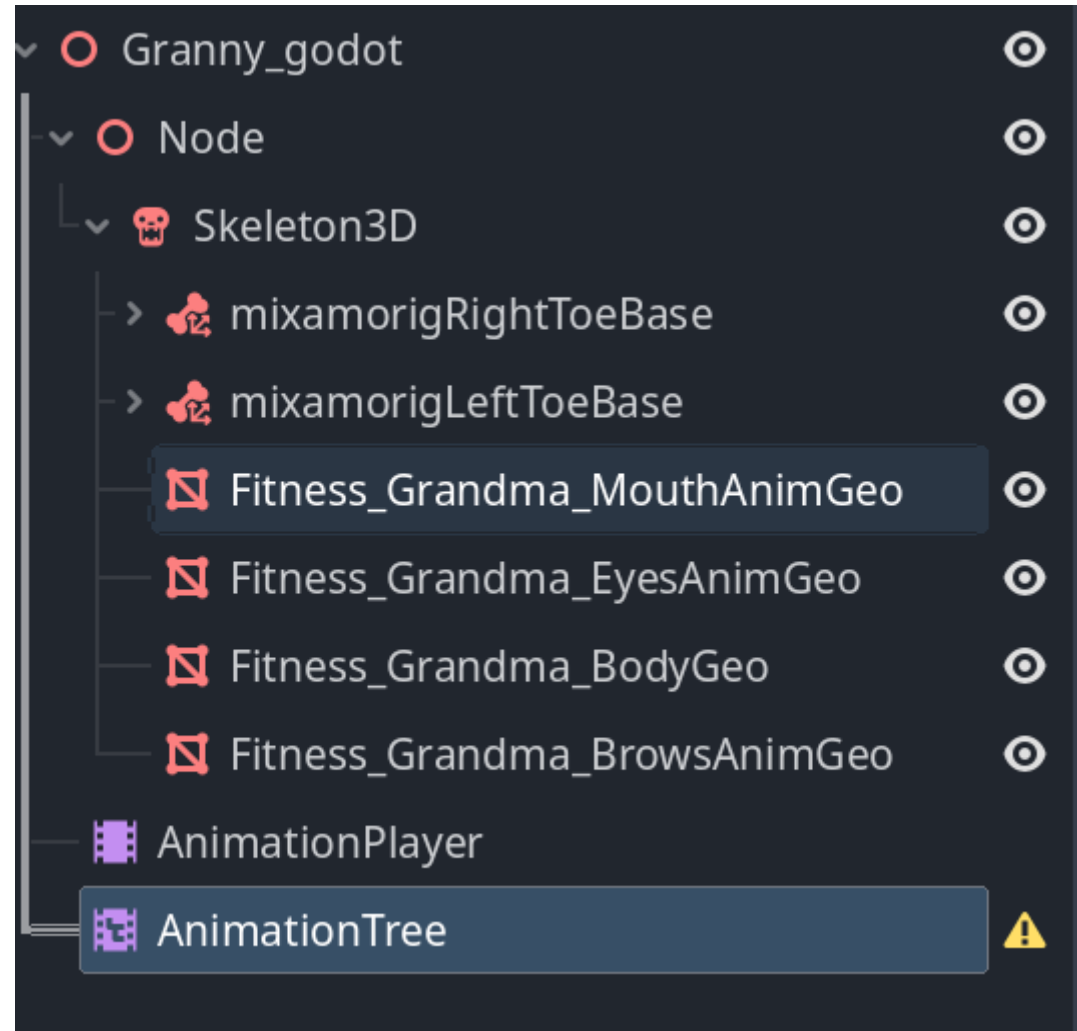
Granny 캐릭터 씬 다시 편집



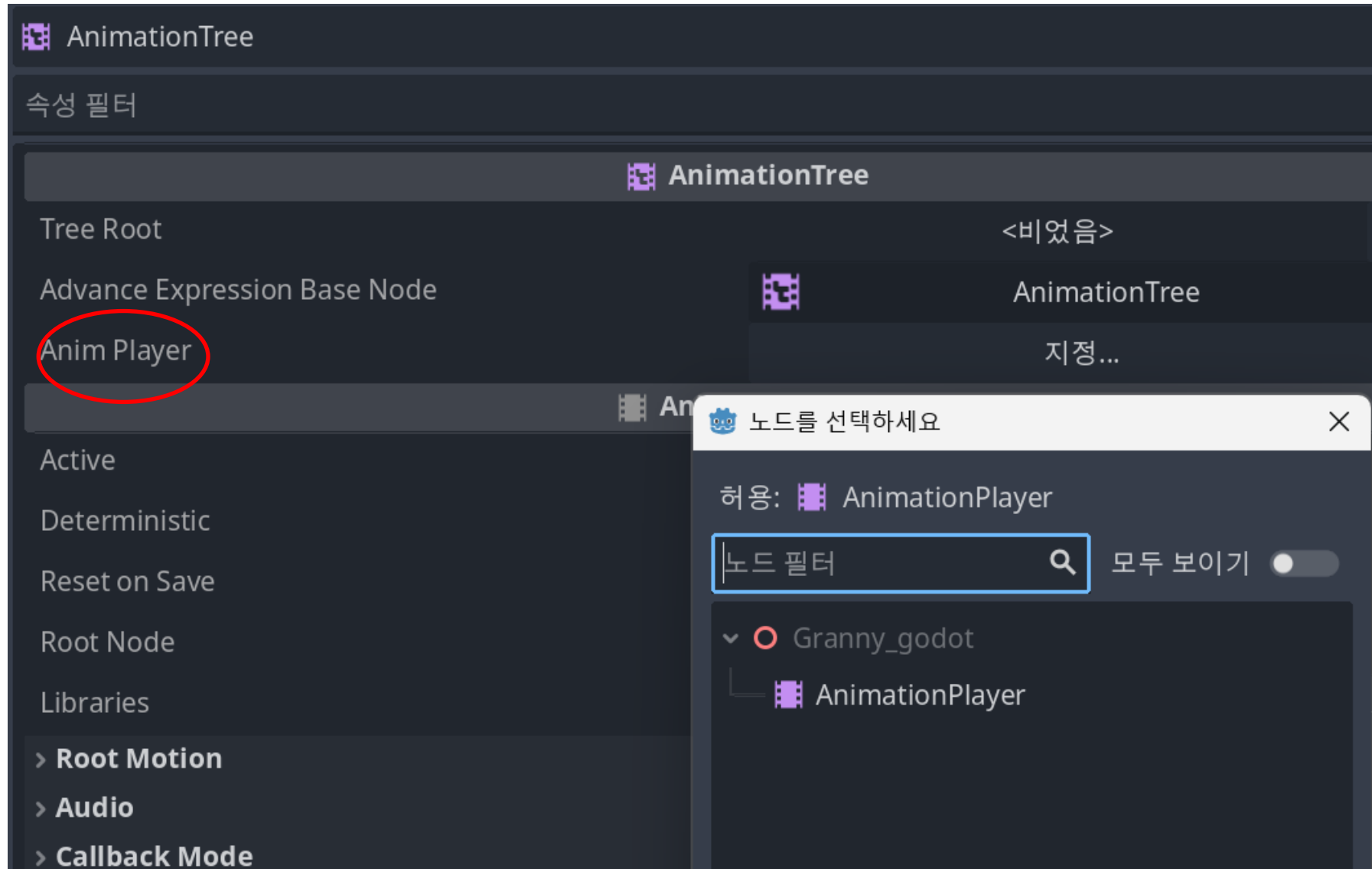
Viewer Scene 만들기



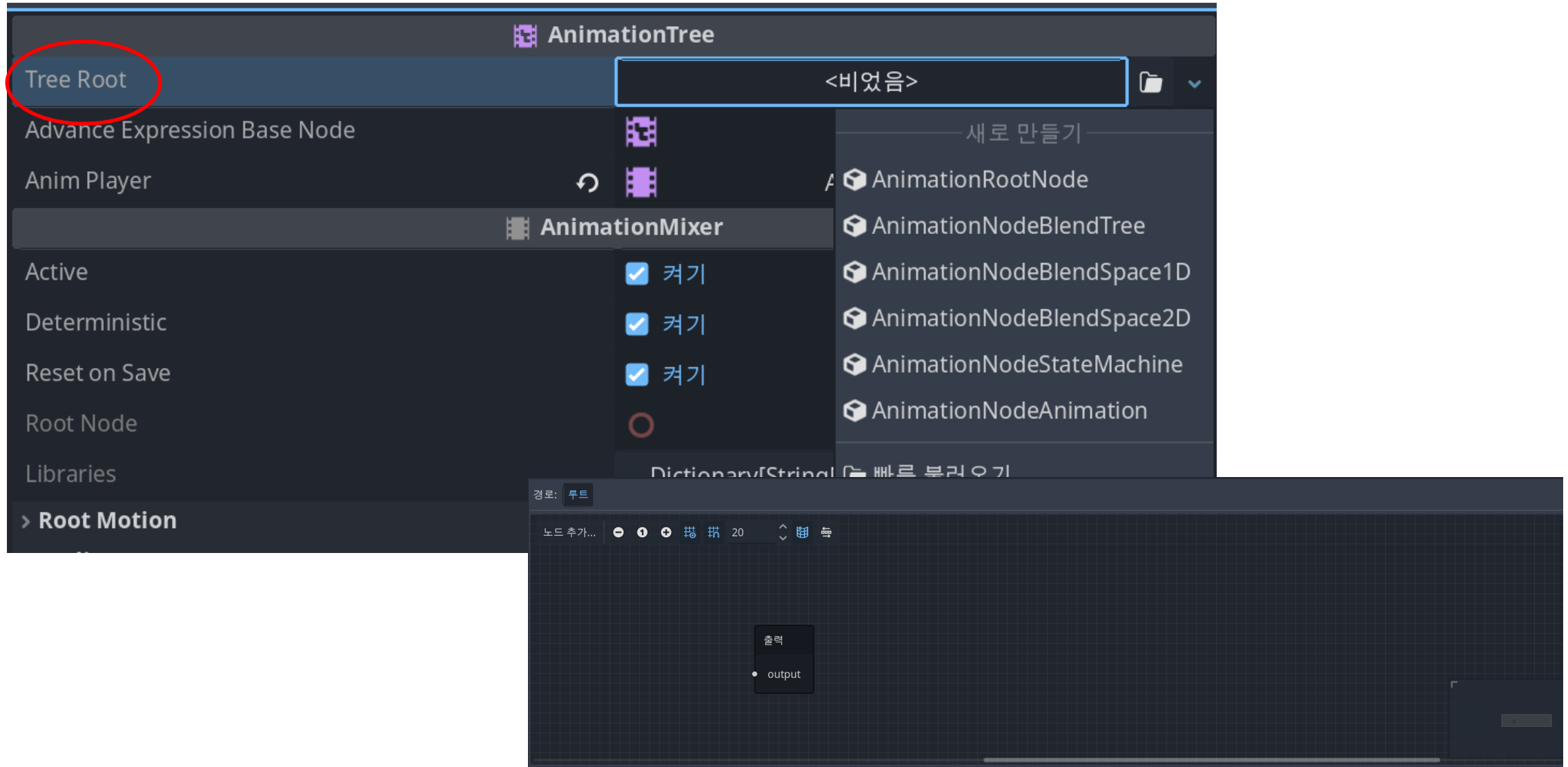
Animation Blending: Animation Tree를 캐릭터 씬에 생성



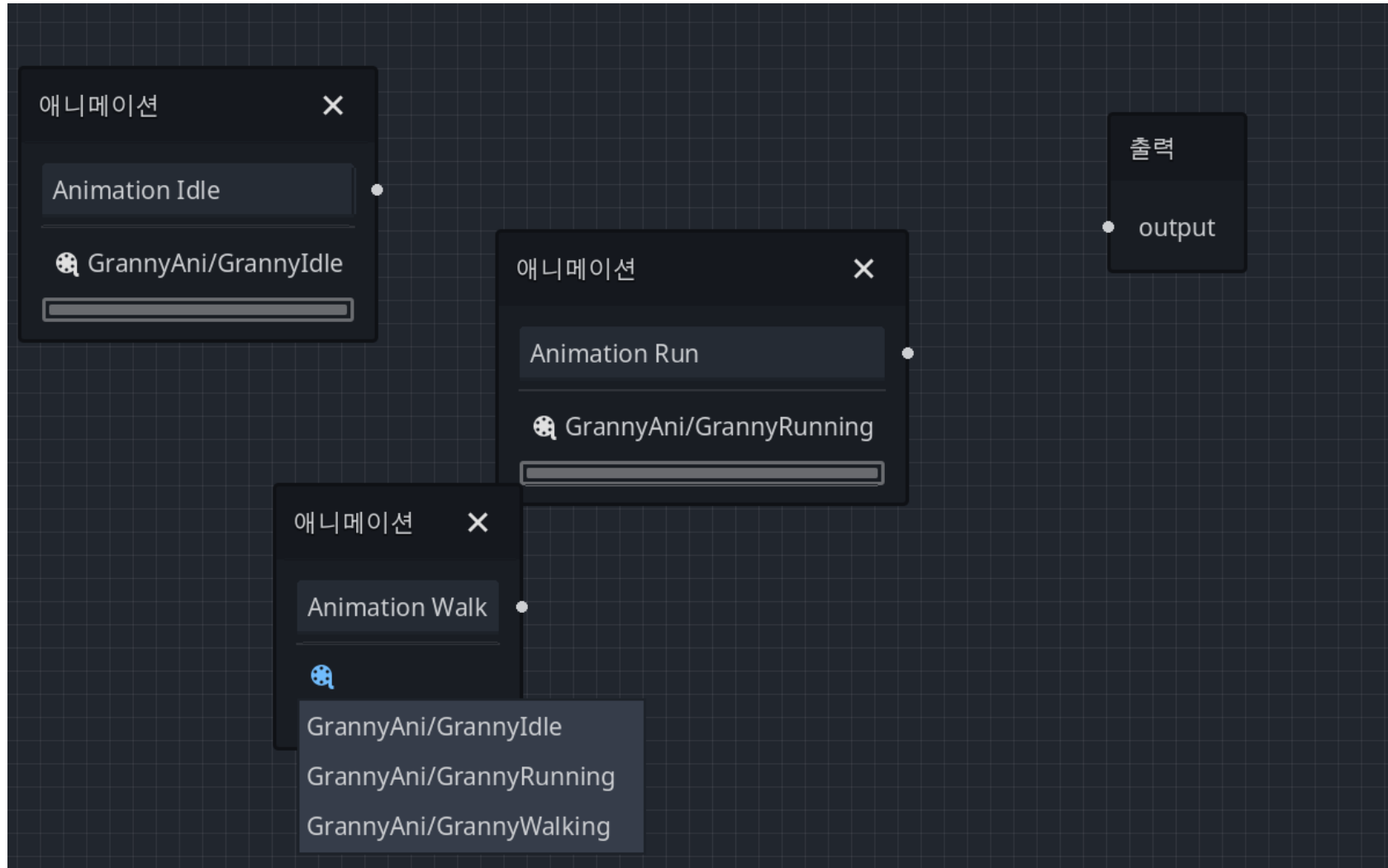
Animation Tree가 사용할 Animation Player 선택



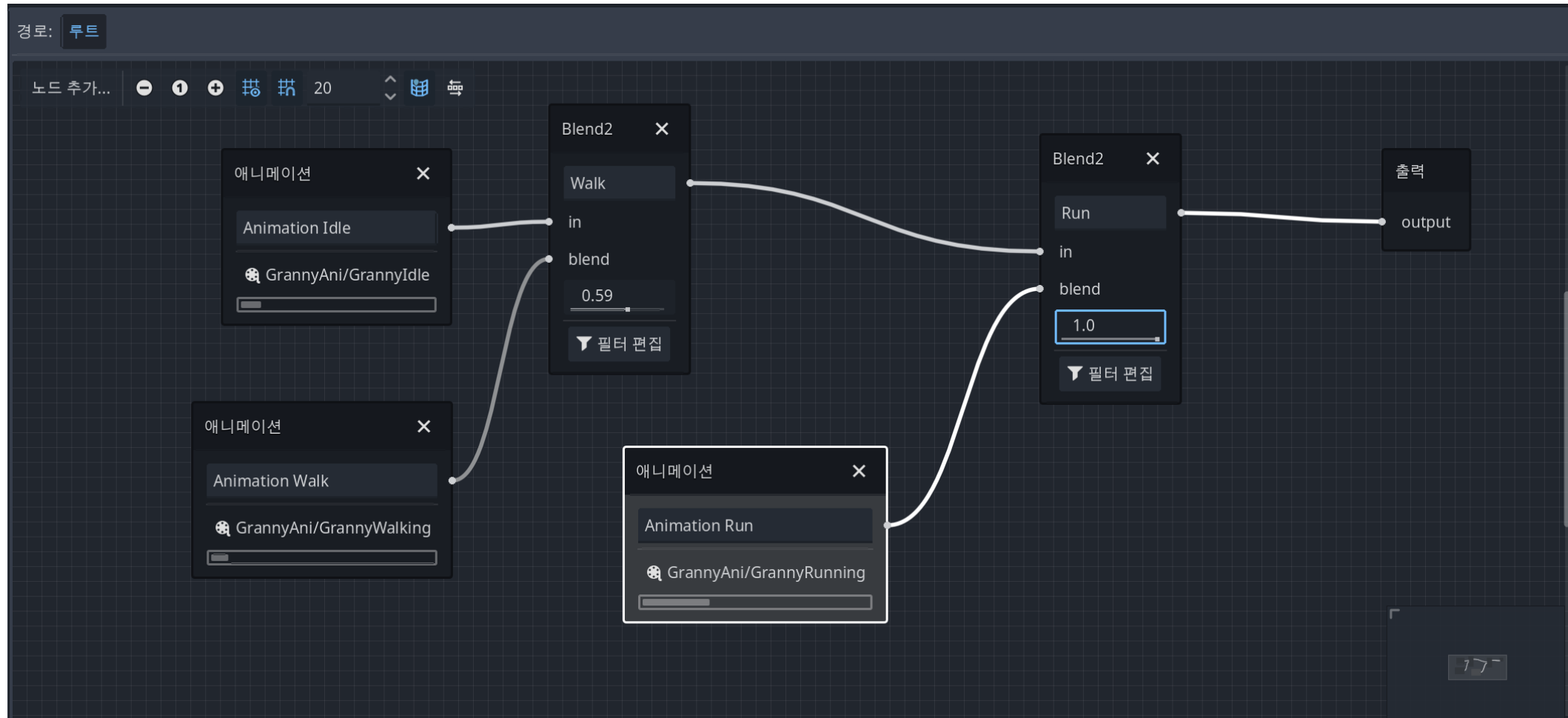
Animation Tree의 루트 설정: AnimationNodeBlendTree



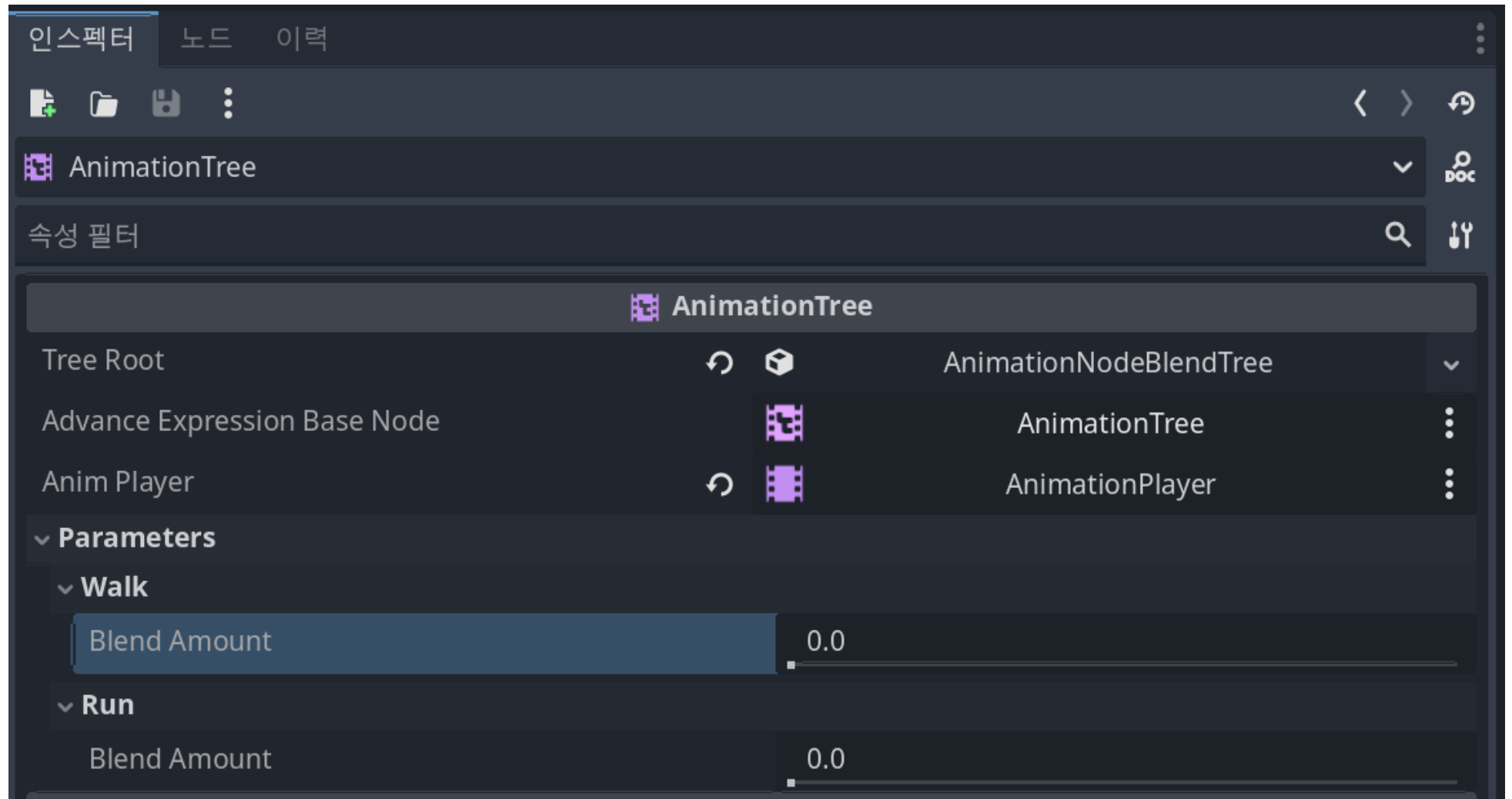
Tree 구성



Blend As You Want



Blending parameters



게임에서 이 동작을 쓰려면? Script로 코딩을 해야... 이걸 다음에

```
@export var blend_speed = 15
```

```
var run_val = 0
```

```
var sit_val = 0
```

```
var jump_val = 0
```

```
func phys|
```

```
func handle_animations(delta):
```

```
    match curAnim:
```

```
        IDLE:
```

```
            run_val = lerpf(run_val,0,blend_speed*delta)
```

```
            sit_val = lerpf(sit_val,0,blend_speed*delta)
```

```
            jump_val = lerpf(jump_val,0,blend_speed*delta)
```

```
        RUN:
```

```
            run_val = lerpf(run_val,1,blend_speed*delta)
```

```
            sit_val = lerpf(sit_val,0,blend_speed*delta)
```

```
            jump_val = lerpf(jump_val,0,blend_speed*delta)
```

```
    if ladder:
```

```
        if input_dir.y == -1:
```

```
            velocity.y = movement_speed * 0.5
```

```
    if direction:
```

```
        # Apply acceleration
```

```
        velocity.x = direction.x * movement_speed
```

```
        velocity.z = direction.z * movement_speed
```

```
        if is_on_floor() and !sitting:
```

```
            curAnim = RUNNING
```

```
    else:
```

```
        velocity.x = move_toward(velocity.x, 0, movement_speed)
```

```
        velocity.z = move_toward(velocity.z, 0, movement_speed)
```

```
        if is_on_floor() and !sitting:
```

```
            curAnim = IDLE
```

```
    # Handle jump.
```

```
    if Input.is_action_pressed("jump") and active:
```

```
        if seat != null:
```

```
            var packet = {"message" : "take_seat", "take" : false, "seat" : se
```



Questions?